Archives of Computational Methods in Engineering (2026) 33:15195–15228
https://doi.org/10.1007/s11831-026-10755-9

REVIEW

A Comprehensive Review on Advanced Deep Computational Techniques for Air Pollution Prediction

Mary Ann Yeboah[1] · Evans Kotei[2] · Thirunavukarasu Ramkumar[3] · Kofi Agyarko Ababio[1] · Maxwell Kwame Boakye[4] · Samuel Dua Oduro[1] · Sampson Kofi Kyei[5]

Received: 15 May 2026 / Revised: 16 July 2026 / Accepted: 27 July 2026 / Published online: 3 August 2026
© The Author(s) 2026

Abstract

Air pollution has become one of the most significant environmental and public health challenges worldwide, necessitating accurate prediction systems to support effective air quality management and informed decision-making. Deep learning has emerged as a powerful approach for modelling the complex nonlinear and spatiotemporal relationships that characterise air quality data. This review provides a comprehensive synthesis of recent advances in deep learning techniques for air pollution prediction. Following the PRISMA 2020 guidelines, the literature was systematically reviewed to evaluate deep learning architectures, environmental data sources, model performance, and emerging research trends. The review traces the evolution of deep learning models from Artificial Neural Networks (ANNs) to Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, Gated Recurrent Units (GRUs), Transformer-based models, transfer learning, and hybrid architectures. It further examines the integration of ground monitoring stations, satellite remote sensing, meteorological observations, IoT-based sensors, and multi-source data fusion for improving prediction accuracy and model robustness. Comparative analysis indicates that hybrid and Transformer-based models consistently achieve superior predictive performance, while RMSE, MAE, and R^2 remain the most widely adopted evaluation metrics. The review also identifies key research challenges, including data scarcity, model interpretability, computational complexity, and benchmark standardisation, and highlights emerging directions such as Explainable Artificial Intelligence, Graph Neural Networks, Physics-Informed Neural Networks, Federated Learning, and Edge AI for developing accurate, scalable, and intelligent air quality prediction systems.

1 Introduction

Air pollution (AP) has become one of the most critical environmental and public health challenges of the twenty-first century. It is characterised by the presence of harmful gases and particulate matter in the atmosphere at concentrations that adversely affect human health, ecosystems, and climate. Pollutants originate from both anthropogenic and natural sources, including vehicular emissions, industrial activities, biomass burning, road dust, agricultural practices, volcanic eruptions, and pollen [1–3]. Rapid industrialisation, urbanisation, population growth, and increasing energy demand have further intensified air pollution levels, making it a global concern that transcends geographical and socioeconomic boundaries.

The health implications of air pollution are profound. According to the World Health Organization (WHO), exposure to ambient air pollution contributes to millions of premature deaths annually and is recognised as one of the

✉ Evans Kotei
evans.kotei@kstu.edu.gh

Thirunavukarasu Ramkumar
ramkumar.thirunavukarasu@vit.ac.in

1 Department of Statistical Sciences, Faculty of Applied Sciences and Technology, Kumasi Technical University, Kumasi, Ghana

2 Department of Computer Science, Faculty of Applied Sciences and Technology, Kumasi Technical University, Kumasi, Ghana

3 Department of Computer Applications, School of Computer Science Engineering and Information Systems, Vellore Institute of Technology, Vellore, Tamil Nadu, India

4 Department of Environmental Science, Faculty of Built and Natural Environment, Ho Technical University, Ho, Ghana

5 Department of Chemical Engineering, Faculty of Engineering and Technology, Kumasi Technical University, Kumasi, Ghana

leading environmental risk factors worldwide [4]. Long-term exposure to polluted air has been associated with numerous diseases, including asthma, chronic obstructive pulmonary disease (COPD), lung cancer, stroke, cardiovascular diseases, and respiratory infections [5]. Beyond its effects on human health, air pollution contributes to climate change, reduces agricultural productivity, degrades ecosystems, and imposes significant economic costs through increased healthcare expenditure and productivity losses. Figure 1 illustrates some of the major health consequences associated with prolonged exposure to air pollution.

To minimise these impacts, governments and environmental agencies continuously monitor ambient air quality using the Air Quality Index (AQI), which provides a standardised measure of pollution levels and their associated health risks. Reliable estimation of pollutants such as particulate matter ($PM_{2.5}$ and PM_{10}), carbon monoxide (CO), sulfur dioxide (SO_2), nitrogen dioxide (NO_2), ozone (O_3), and lead (Pb) is essential for environmental regulation, public health protection, and sustainable urban planning [6]. Accurate and forecasting of these pollutants enable authorities to issue timely warnings, implement pollution control measures, and formulate evidence-based environmental policies.

1.1 Air Quality Monitoring and Existing Challenges

Conventional air quality monitoring relies primarily on fixed-site monitoring stations equipped with regulatory-grade instruments, high-volume and low-volume samplers, and passive sampling systems. Although these methods provide highly accurate pollutant measurements, they suffer from several practical limitations, including high installation and maintenance costs, limited geographical coverage, and inadequate temporal resolution, particularly in low- and middle-income countries [7]. Consequently, many regions lack sufficient monitoring infrastructure to support continuous and comprehensive air quality assessment.

Recent advances in sensing technologies have expanded the availability of alternative monitoring approaches, including low-cost sensor networks, satellite remote sensing, mobile sensing platforms, Internet of Things (IoT) devices, and meteorological observation systems. These technologies provide broader spatial coverage, lower deployment costs, and near real-time measurements, making them attractive

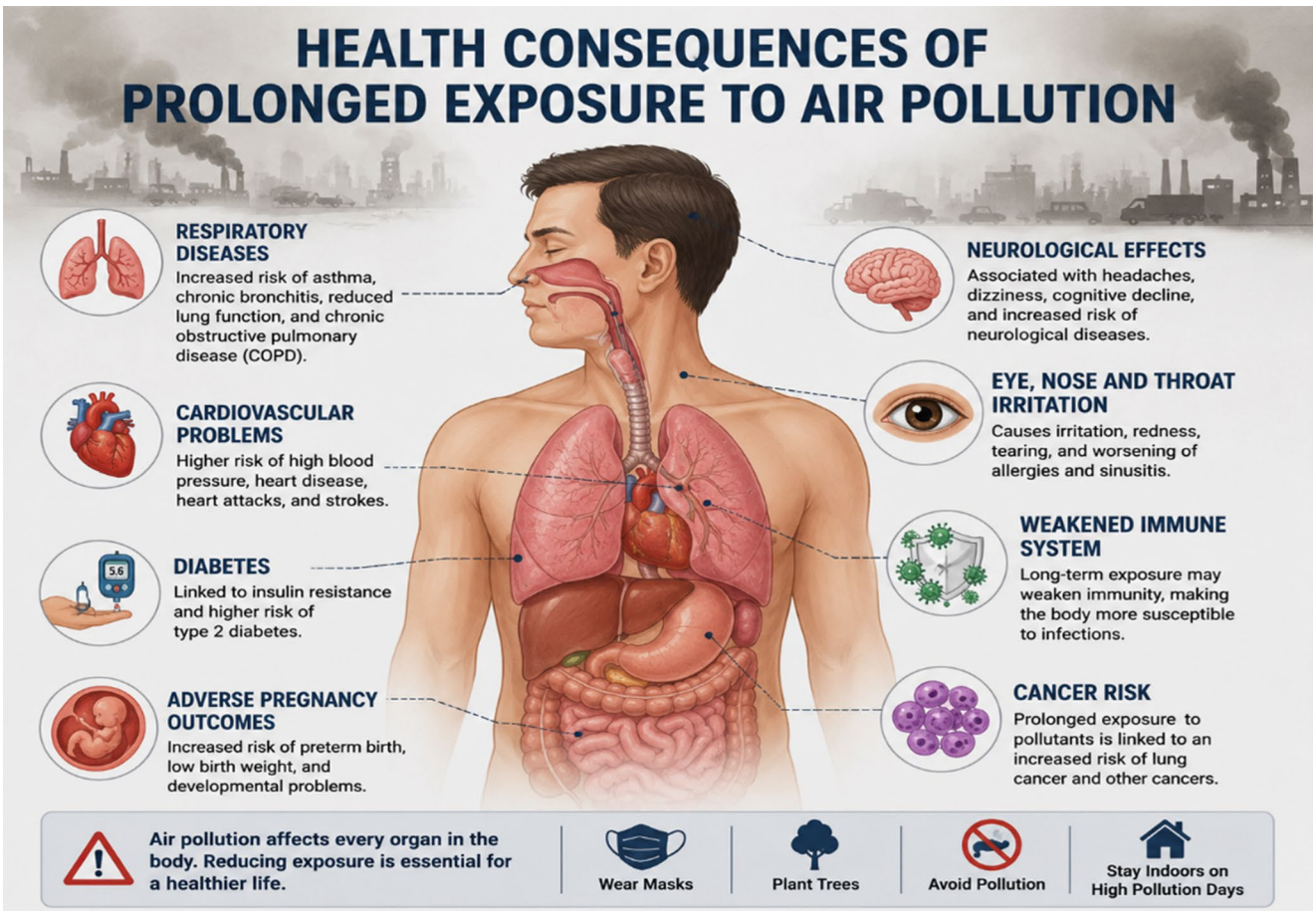


Fig. 1 Consequences of prolonged exposure to air pollution

complements to traditional monitoring networks [8]. Figure 2 presents a simplified architecture of contemporary air pollution measurement systems.

Among these technologies, low-cost sensor networks and satellite remote sensing have emerged as two of the most widely adopted approaches. Low-cost sensors enable dense monitoring networks because of their affordability and ease of deployment, whereas satellite observations provide extensive geographical coverage and long-term environmental records that are particularly valuable in regions with limited ground-based monitoring infrastructure [9, 10]. Satellite-derived measurements have been successfully used to estimate $PM_{2.5}$ concentrations across Africa and Asia, where conventional monitoring stations remain sparse [7, 10–12].

Despite these advances, several challenges continue to hinder effective air quality monitoring. Measurements from low-cost sensors often require calibration and compensation for environmental factors such as temperature and humidity. Satellite observations may be affected by cloud cover, atmospheric interference, and retrieval uncertainties. Furthermore, sensor failures, communication interruptions, and maintenance constraints frequently result in incomplete or missing datasets, reducing the reliability of predictive models. Air pollutant concentrations are also governed by highly nonlinear interactions among meteorological variables, emission sources, land use, topography, and human activities, making accurate modelling particularly challenging. Traditional statistical approaches and conventional machine learning algorithms often struggle to capture these complex spatial and temporal dependencies and frequently exhibit poor generalisation when applied across different geographical regions or climatic conditions.

1.2 Deep Learning for Air Pollution Prediction

The increasing availability of large-scale environmental datasets has accelerated the adoption of deep learning (DL) techniques for air quality monitoring. Unlike traditional statistical models and shallow machine learning algorithms, deep learning models automatically learn hierarchical feature representations from heterogeneous data sources, enabling them to capture complex nonlinear relationships without extensive manual feature engineering.

Different deep learning architectures have demonstrated strengths for specific air pollution applications. Convolutional Neural Networks (CNNs) have shown exceptional capability in extracting spatial information from satellite imagery and remote sensing data. Conversely, Recurrent Neural Networks (RNNs), especially Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks, have become the dominant approaches for modelling temporal dependencies in sequential air quality data, leading to substantial improvements in forecasting pollutant concentrations [13].

More recently, advances in hybrid deep learning architectures, attention mechanisms, and Transformer-based models have significantly enhanced prediction accuracy by simultaneously capturing spatial and temporal relationships within environmental datasets [14]. In addition, transfer learning has emerged as an effective solution for

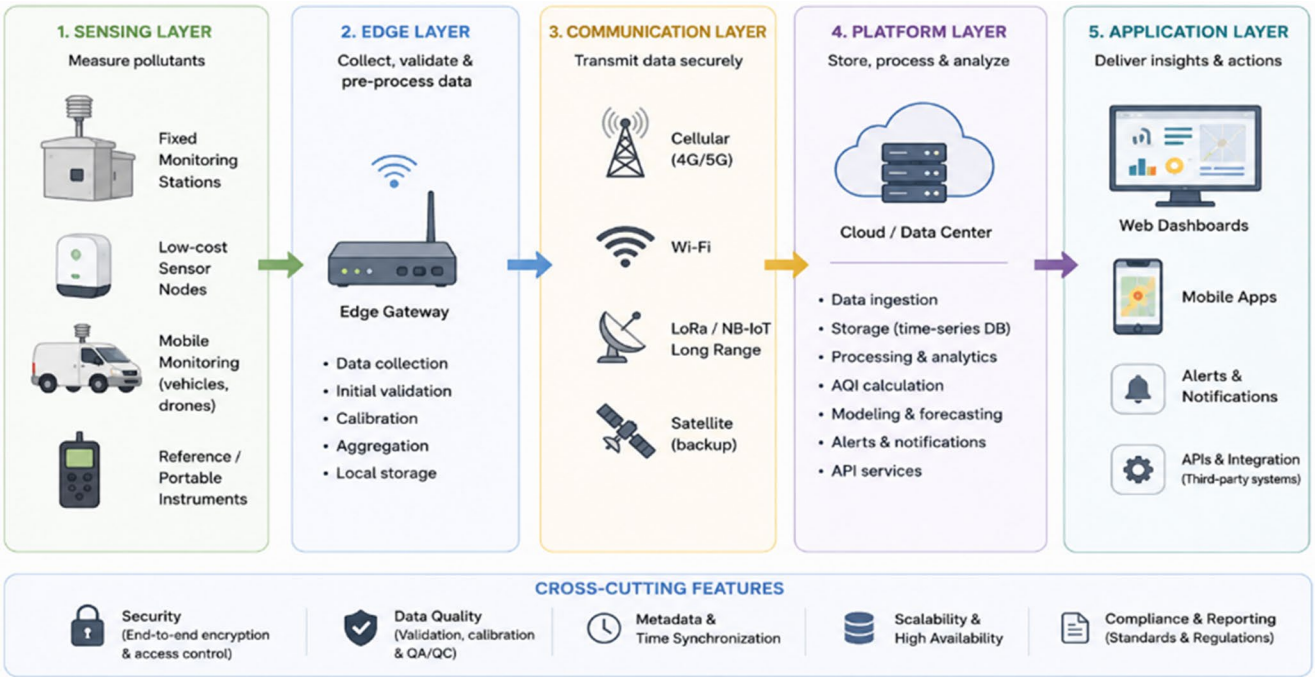


Fig. 2 Simplified architecture of contemporary air pollution measurement system

improving model performance in situations where labelled environmental data are limited. By transferring knowledge from pre-trained models to new prediction tasks, transfer learning reduces overfitting while improving generalisation across diverse environmental conditions [15].

1.3 Research Gap

Although deep learning has attracted considerable attention for air pollution monitoring during the past decade, the existing literature remains fragmented. Most published studies concentrate on individual deep learning algorithms or specific prediction tasks, with considerably less attention devoted to pollutant comparative model evaluation, and operational deployment. Existing review articles generally discuss deep learning applications from isolated perspectives, focusing either on particular pollutants, individual architectures, or limited application domains. Consequently, there remains a lack of a comprehensive review that systematically compares deep learning models, environmental datasets, evaluation metrics, application areas, and practical implementation challenges within a unified analytical framework.

Furthermore, limited emphasis has been placed on understanding how different data sources influence predictive performance, how emerging hybrid and attention-based models compare with conventional architectures, and what challenges must be addressed before deep learning can be widely adopted for intelligent air quality management.

1.4 Objectives and Contributions

Motivated by these research gaps, this review provides a comprehensive assessment of deep learning techniques for air pollution forecasting and prediction. The review critically analyses recent developments in deep learning architectures, examines the characteristics of commonly used environmental datasets, evaluates model performance using widely adopted assessment metrics, and identifies current research challenges together with emerging opportunities for future development.

The major contributions of this review are summarised as follows:

- To systematically review deep learning models developed for air pollution forecasting and prediction.
- To examine the characteristics of environmental datasets, including ground-based monitoring stations, satellite imagery, meteorological observations, IoT sensors, and geographic information systems.
- To compare the predictive performance, robustness, interpretability, and computational characteristics of CNN, LSTM, GRU, Transformer, attention-based, transfer learning, and hybrid deep learning models.
- To identify current research challenges, methodological limitations, and practical deployment issues affecting intelligent air quality monitoring systems.
- To highlight emerging research directions and provide recommendations for future deep learning applications in sustainable air quality management.

1.5 Research Questions

To achieve these objectives, this review is guided by the following research questions:

RQ1 What deep learning models have been proposed for the forecasting, and prediction of air pollution, and how have these models evolved in recent years?

RQ2 What environmental data sources are commonly employed in deep learning-based air pollution studies, and how do these datasets influence model performance?

RQ3 How do different deep learning architectures, including CNN, LSTM, GRU, Transformer, attention-based, and hybrid models, compare in terms of predictive accuracy, robustness, interpretability, and computational efficiency?

RQ4 What are the major challenges, research gaps, and future research opportunities for applying deep learning techniques to operational air quality monitoring and management?

The remainder of this paper is organised as follows. Section 2 presents the review methodology, including the literature search strategy, study selection process, and quality assessment. Sections 3–6 synthesise the existing literature by addressing each research question through analyses of deep learning architectures, environmental datasets, comparative model performance, and current research challenges. Finally, Sect. 7 concludes the review by summarising the key findings and outlining promising directions for future research. Figure 3 show a graphical flow of the study.

2 Review Methodology

This review adopts a systematic literature review (SLR) methodology to provide a comprehensive and unbiased synthesis of recent advances in deep learning techniques for air pollution forecasting and prediction. The review protocol was designed to ensure transparency, reproducibility, and comprehensive coverage of the existing literature while

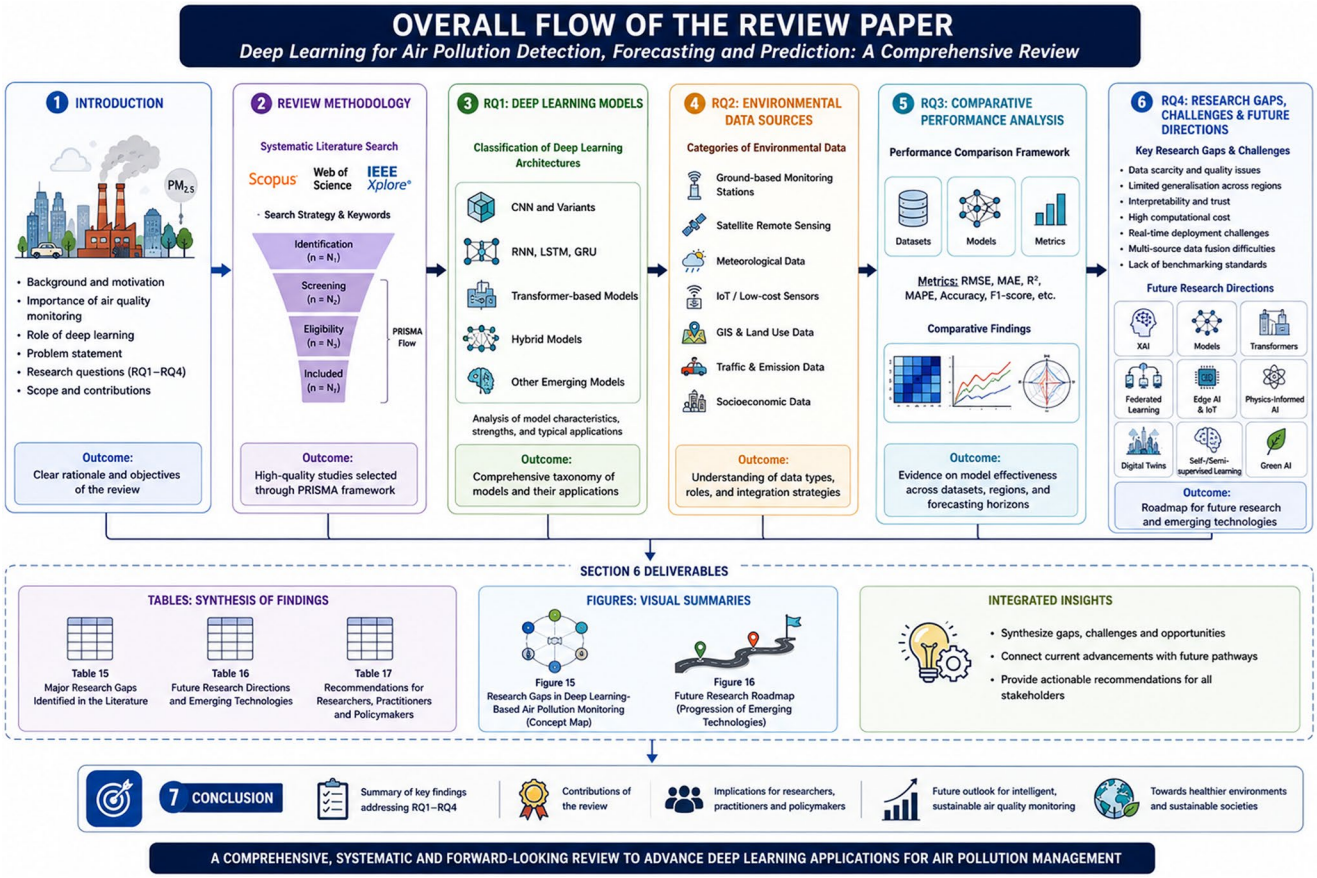


Fig. 3 Graphical flow of the study

minimising selection bias. The methodology consists of five major stages: (i) formulation of the review objectives and research questions, (ii) literature search, (iii) study selection, (iv) quality assessment and data extraction, and (v) synthesis and analysis of the selected studies.

2.1 Review Framework

The review was guided by the four research questions presented in Sect. 1, which define the scope of the study and provide the analytical framework for synthesising the literature. The approach enables a more critical comparison of existing studies and facilitates the identification of emerging trends, methodological strengths, and unresolved research challenges.

2.2 Literature Search Strategy

A comprehensive literature search was conducted using several internationally recognised scientific databases, including Scopus, Web of Science, IEEE Xplore, ScienceDirect, SpringerLink and Google Scholar. These databases were selected because they index high-quality peer-reviewed publications in artificial intelligence, environmental science, remote sensing, atmospheric science, and data analytics. The search focused on journal articles, conference papers, and books published in English between 2018 and 2026, reflecting the rapid evolution of deep learning techniques during this period. To maximise coverage, combinations of keywords and Boolean operators were employed to query the selected databases and the results are in Table 1.

Additional studies were identified through backward and forward snowballing by examining the reference lists and citation networks of highly relevant publications.

2.3 Eligibility Criteria

To ensure the inclusion of high-quality and relevant studies, predefined inclusion and exclusion criteria were established before the literature search. The inclusion and exclusion criteria are in Table 2.

Table 1 Literature search from databases and search strategy

Database	Coverage	Search keywords	Output
Scopus	Peer-reviewed journals and conference proceedings	"Transformer" AND "Air Pollution" "Hybrid Deep Learning" AND "Air Quality"	152
Web of Science	Multidisciplinary research	"Air Pollution" AND "Deep Learning"	33
IEEE Xplore	Engineering and AI	"Air Pollution " AND "CNN"	78
ScienceDirect	Environmental sciences	"Air Quality Forecasting" AND "GRU"	29
SpringerLink	AI and environmental science	"$PM_{2.5}$ Prediction" AND "Neural Networks"	76
Google Scholar	Computing research and supplementary research	"Satellite Imagery" AND "Air Pollution"	41
Web of Science	Computing research	"Air Quality Prediction" AND "LSTM"	34
Snowballing approach	Computing research		27

Table 2 Inclusion and exclusion criteria

Criterion	Inclusion	Exclusion
Language	English	Non-English
Publication Type	Journal articles and conference papers	Editorials, theses, book chapters, reports
Publication Period	2018–2026	Before 2018 (unless seminal)
Methodology	Deep Learning	Conventional statistical methods only
Application	Forecasting, Prediction	Other environmental applications
Dataset	Air pollution datasets	Unrelated datasets
Evaluation	Quantitative performance metrics	No experimental evaluation

2.4 Study Selection Process

The study selection process followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines to ensure transparency and reproducibility.

Initially, all records retrieved from the selected databases were imported into a reference management system, where duplicate publications were removed. The remaining studies underwent title and abstract screening to eliminate publications that were outside the scope of this review. The full texts of potentially eligible articles were subsequently assessed against the predefined inclusion and exclusion criteria. Studies meeting all eligibility requirements were retained for detailed qualitative analysis.

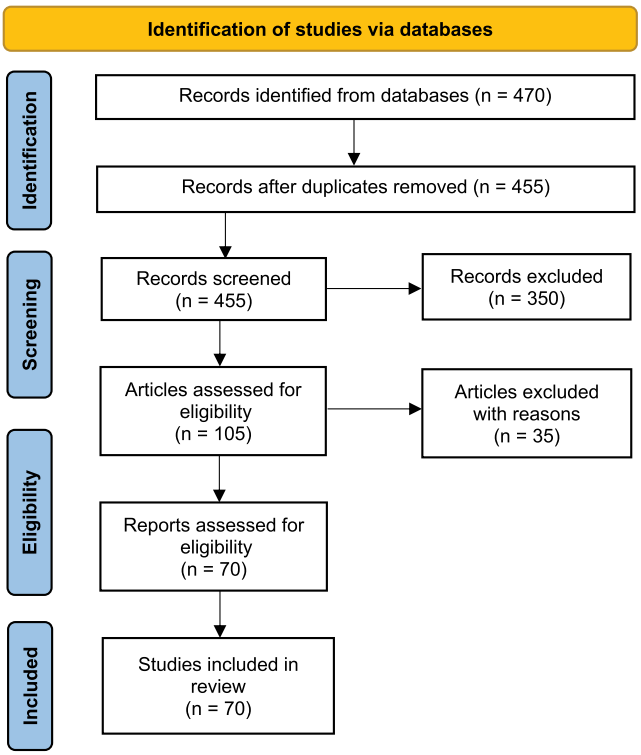


Fig. 4 PRISMA flow diagram

The complete study selection procedure is illustrated using the PRISMA flow diagram shown in Fig. 4.

2.5 Data Synthesis and Analysis

The selected studies were synthesised using a thematic analysis aligned with the four research questions. First, the reviewed literature was analysed to identify the evolution of deep learning architectures for air pollution applications (RQ1). Subsequently, commonly used environmental datasets and their influence on model performance were examined (RQ2). The comparative strengths and weaknesses of different deep learning models were then evaluated using commonly reported performance metrics such as Accuracy, Precision, Recall, F1-score, Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), and the coefficient of determination (R^2) (RQ3). Finally, existing limitations, deployment challenges, and future research opportunities were synthesised to address RQ4.

2.6 Bibliometric Findings

To complement the systematic literature review, a bibliometric analysis was conducted to provide a quantitative overview of research trends in deep learning applications for air pollution forecasting and prediction.

The co-occurrence of keywords in the reviewed literature was examined using **VOSviewer** software (version 1.16.0). As shown in Fig. 5, the analysis was based on a total of **899 keywords**, with only those appearing **at least 10 times** included in the final visualisation. This threshold resulted in the identification of **23 distinct keywords**. In the generated map, links between keywords represent their frequent co-occurrence within the same documents, while **thematic similarities** are visualised through **cluster groupings**, indicating conceptual relationships among the terms. Three clusters were identified, and given thematic titles, cluster one being Deep Learning and AI techniques, cluster two being Air Pollution and variables and cluster three being Predictive Modelling. In particular, terms such as deep learning, air pollution, air quality, particulate matter, atmospheric pollution, long short-term memory, learning systems, and forecasting appeared most frequently, as indicated by the larger ball.

The analysis was performed using the final set of publications selected through the systematic review process. Information extracted from each publication included publication year, country of origin, journal source, author keywords, and citation metadata. These data were synthesised to examine publication trends, geographical distribution of research, leading publication venues, and keyword co-occurrence patterns.

2.6.1 Country Contributions

Research on deep learning for air pollution monitoring is geographically diverse, although publication output remains concentrated in a relatively small number of countries.

Figure 6 shows the countries that are collaborating with other countries in researching into deep leaning techniques used in detecting and predicting air pollution. This is based on a minimum of three documents per country. It is obvious that authors from India are co-authoring with other countries the most, with China and the United States following respectively.

These countries have benefited from extensive environmental monitoring infrastructures, publicly available air quality datasets, and active collaborations between academia, government agencies, and industry. In contrast, relatively few studies originate from Africa and other developing regions, primarily because of limited monitoring infrastructure, data scarcity, and restricted research funding. This imbalance highlights an important opportunity for future research aimed at improving air quality monitoring in underrepresented regions.

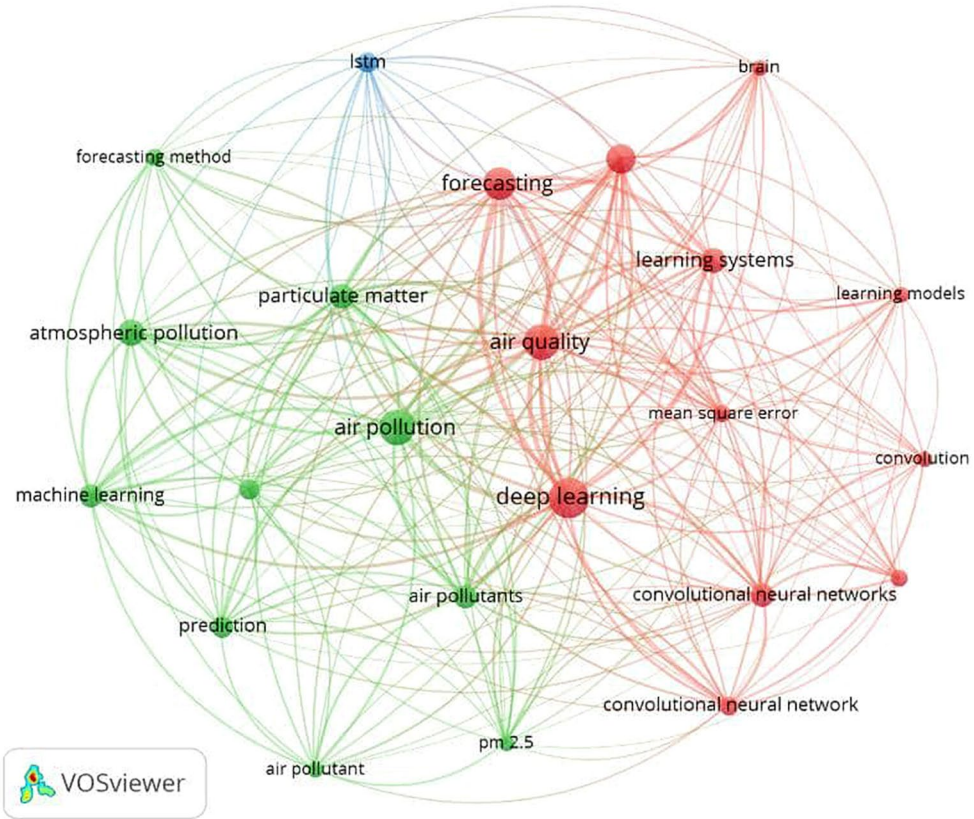


Fig. 5 Network of keyword co-occurrence in literature based on deep leaning

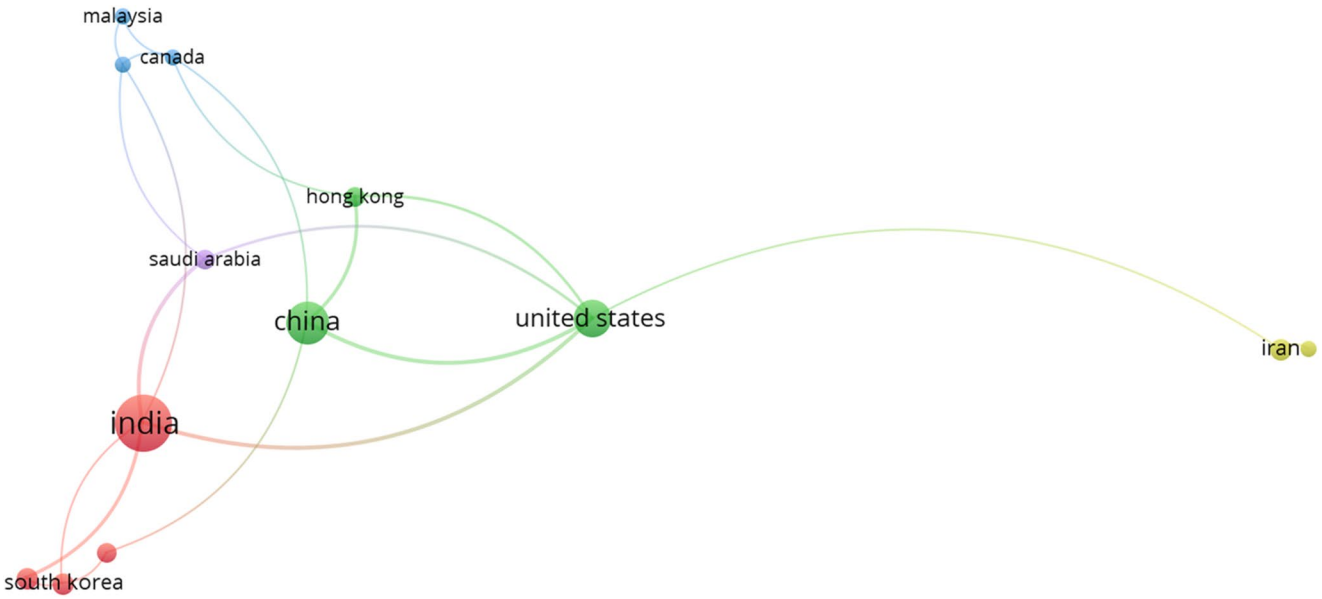


Fig. 6 Co-authorship by country network in literature based on Deep leaning

2.6.2 Leading Publication Venues

The selected studies were published across a wide range of journals spanning environmental science, atmospheric research, artificial intelligence, remote sensing, and environmental informatics. The multidisciplinary nature of this research area reflects the integration of environmental monitoring with advanced computational techniques. Several journals have consistently published significant numbers of studies on deep learning for air pollution monitoring, including: Atmospheric Environment, Science of the Total Environment, Environmental Pollution Environmental Modelling & Software, Remote Sensing, IEEE Access, Expert Systems with Applications, Environmental Research, Atmospheric Pollution Research, Applied Soft Computing. The growing presence of studies in artificial intelligence and data science journals further demonstrates the increasing convergence of environmental science and machine learning research.

2.6.3 Bibliometric Insights

The bibliometric analysis reveals several important observations:

1. Research output has grown rapidly since 2018, reflecting increasing interest in deep learning-based air quality monitoring.
2. India, China and the United States are the leading contributors to the field, while studies from developing regions remain comparatively limited.
3. Environmental science and artificial intelligence journals jointly dominate publication output, demonstrating the interdisciplinary nature of this research area.
4. Keyword analysis indicates a transition from conventional neural networks towards hybrid architectures, attention mechanisms, and Transformer-based models.
5. Emerging research themes increasingly emphasise explainable artificial intelligence, graph neural networks, multimodal data fusion, and intelligent environmental monitoring systems.

These findings provide valuable context for the thematic analysis presented in the following sections, where the literature is synthesised according to the four research questions developed in this review.

2.7 Summary of the Methodology

The adopted systematic review methodology ensures that the evidence presented in this study is comprehensive, transparent, and reproducible. By integrating a rigorous literature search strategy, predefined eligibility criteria, structured quality assessment, and research question-driven thematic synthesis, the review provides a robust foundation for critically evaluating recent advances in deep learning-based air pollution forecasting and prediction. The subsequent sections present the findings according to the four research questions introduced in Sect. 1, beginning with an examination of the evolution of deep learning models for air pollution applications.

3 Deep Learning Models for Air Pollution Forecasting and Prediction (RQ1)

Deep learning has revolutionised air quality monitoring by enabling the automatic extraction of complex nonlinear relationships from heterogeneous environmental datasets. Unlike traditional statistical methods and shallow machine learning algorithms that rely heavily on handcrafted feature engineering [16], deep learning models learn hierarchical feature representations directly from raw data, making them particularly suitable for modelling the intricate spatial and temporal dynamics of atmospheric pollution.

The evolution of deep learning for air pollution applications has closely followed developments in artificial intelligence. Early studies primarily employed Artificial Neural Networks (ANNs) and shallow multilayer perceptron for pollutant concentration prediction [17]. However, these models exhibited limited capability in capturing complex nonlinear relationships and long-term temporal dependencies. As environmental datasets became increasingly large and diverse, more sophisticated deep learning architectures including Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, Gated Recurrent Units (GRUs), Transformer models, attention mechanisms, transfer learning, and hybrid architectures were introduced to overcome these limitations.

3.1 Evolution of Deep Learning Architectures

The development of deep learning techniques for air pollution monitoring can be broadly categorised into four successive stages. The first stage relied on shallow neural networks and conventional machine learning algorithms that utilised manually engineered features extracted from meteorological observations and historical pollutant concentrations [18]. Although these models provided reasonable predictive capability, they generally failed to capture the complex nonlinear interactions between atmospheric variables. Figure 7 illustrates the evolution of deep learning architectures for air pollution applications.

The second stage witnessed the adoption of Convolutional Neural Networks (CNNs), which significantly improved spatial feature extraction from satellite imagery, aerosol optical depth maps, and multisource environmental datasets. CNNs demonstrated superior performance in identifying pollution hotspots, estimating particulate matter distributions, and detecting emission sources due to their hierarchical spatial representation learning [19].

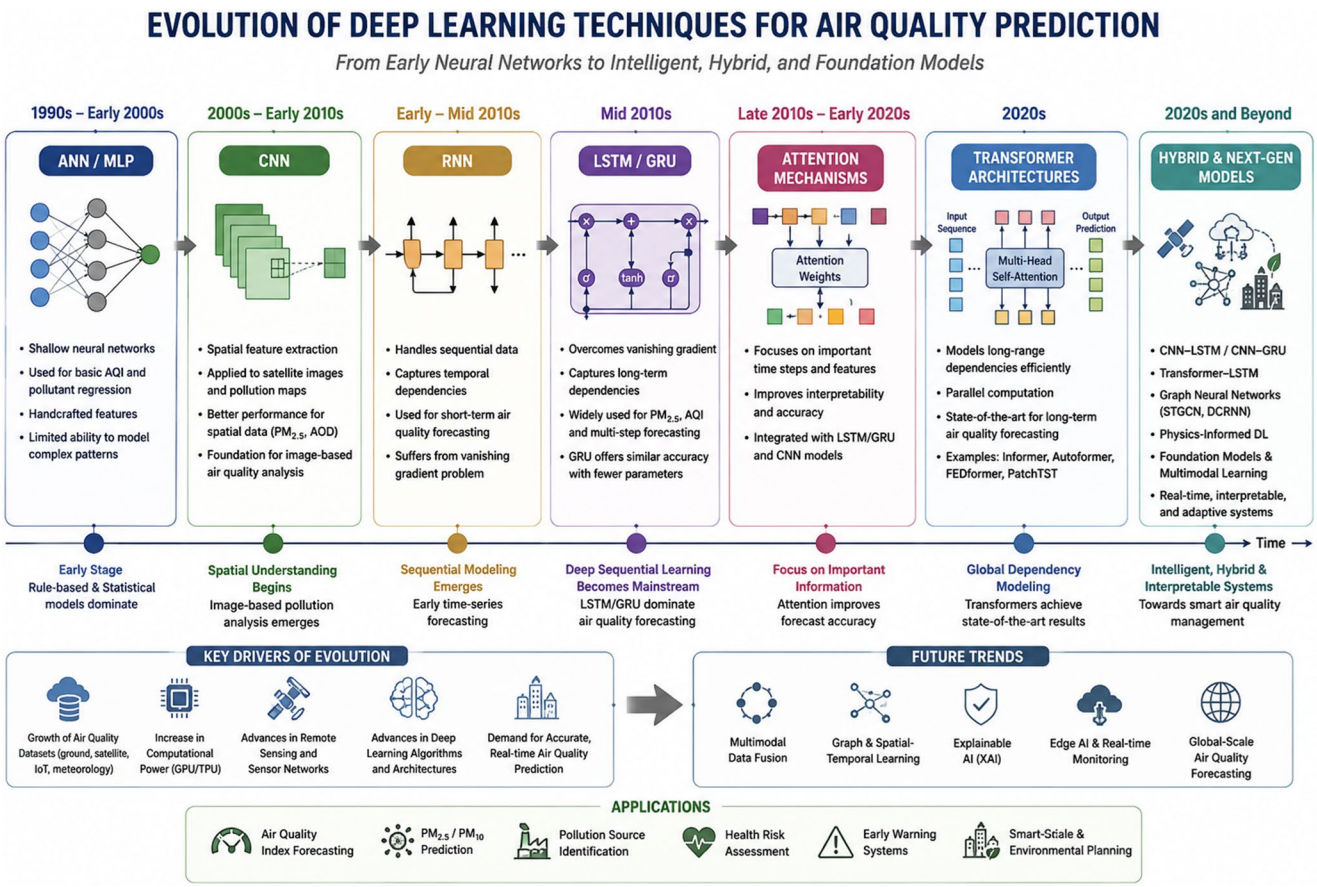


Fig. 7 Evolution of deep learning architectures air quality prediction

The third stage introduced recurrent architectures, particularly Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU) models. These architectures addressed the limitations of CNNs in modelling sequential information by learning temporal dependencies from historical air quality observations. Consequently, LSTM and GRU rapidly became the dominant approaches for short-term and long-term air quality forecasting [20, 21].

More recently, the emergence of Transformer architectures, attention mechanisms, graph neural networks, and hybrid deep learning models has substantially improved prediction accuracy by integrating both spatial and temporal information. These advanced models can simultaneously exploit heterogeneous datasets, including satellite imagery, meteorological variables, traffic information, IoT sensor networks, and geographical information systems, thereby providing more accurate and robust pollution forecasts.

3.2 Convolutional Neural Networks for Spatial Air Pollution Prediction

Convolutional Neural Networks (CNNs) represent one of the earliest deep learning architectures successfully applied to environmental monitoring. Originally developed for computer vision tasks, CNNs automatically learn hierarchical spatial features from images through convolutional operations, pooling layers, and nonlinear activation functions. Figure 8 shows a generic CNN model for classification.

Within air pollution research, CNNs are predominantly employed for spatial analysis rather than temporal prediction. Their ability to extract local spatial patterns makes them particularly suitable for analysing satellite imagery, aerosol optical depth products, land-use maps, traffic density maps, and multispectral remote sensing datasets. These capabilities have enabled researchers to identify pollution hotspots, estimate particulate matter concentrations, detect emission sources, and generate high-resolution pollution maps.

Several studies have demonstrated that CNN-based models consistently outperform conventional machine learning techniques for image-based pollution because they eliminate the need for manual feature extraction while effectively learning spatial relationships among neighbouring regions. Furthermore, transfer learning has enabled pre-trained CNN models to be adapted to new geographical regions where labelled environmental datasets are limited, reducing computational cost and improving generalisation.

A small-sample CNN was used in [22] in combination with computational fluid dynamics (CFD) simulations

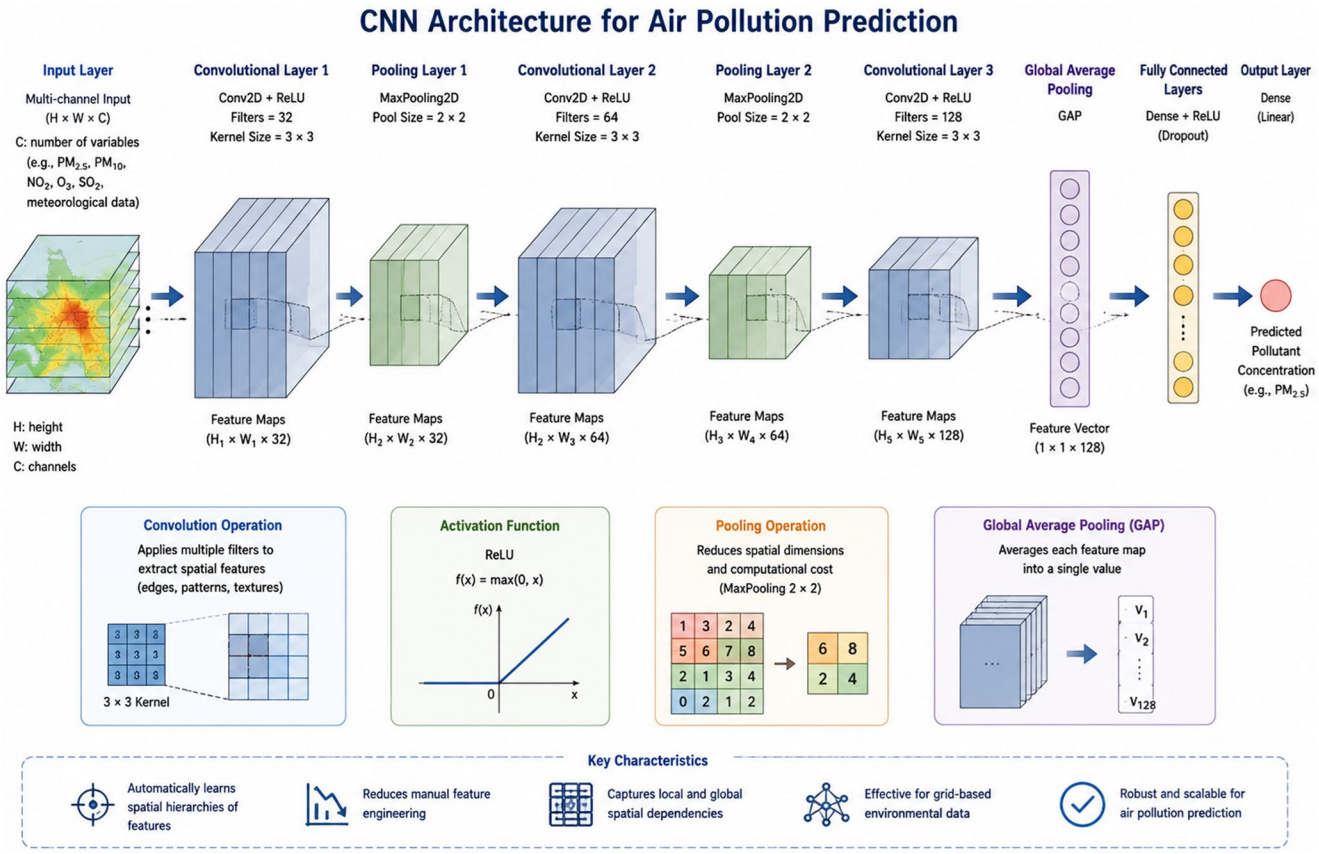


Fig. 8 Network architecture of CNN

and multi-scenario experimental validation, successfully achieved effective and inversion of large-scale pollution sources. The model attained a weighted average (WA) accuracy of 82.3% within a 1 m prediction radius, with a Normalised Euclidean Distance (NED) of less than 0.26%. These results demonstrate the model's capability to accurately locate pollution sources within a quite small spatial range.

The work proposed in [23] explores the use of CNN to predict air pollution concentrations, specifically estimating PM_{10} levels for the next 24 h. Data collected from an Airly sensor station in a village near Kraków were employed to train and evaluate the network. The study's results confirm the strong effectiveness of neural networks in forecasting PM_{10} levels. When compared with a linear model, both standard CNN and time-aware CNN models delivered markedly better performance, highlighting the advantage of CNN-based techniques in air quality prediction.

Despite these advantages, CNNs exhibit several limitations. Their receptive fields primarily capture spatial information, making them less effective for modelling long-term temporal dependencies in sequential air quality observations. Consequently, CNNs are often integrated with recurrent architectures to jointly model spatial and temporal characteristics.

Overall, CNNs remain the preferred architecture for pollution source localisation, satellite image interpretation, aerosol optical depth estimation, and spatial distribution modelling.

3.3 Long Short-Term Memory Networks

Long Short-Term Memory (LSTM) networks represent the most extensively adopted deep learning architecture for air pollution forecasting [24]. LSTMs extend conventional RNNs by incorporating memory cells and gating mechanisms that regulate information flow, enabling them to preserve relevant historical information over extended time periods while discarding irrelevant observations [25].

The ability of LSTM networks to capture long-term temporal dependencies has made them particularly effective for predicting concentrations of $PM_{2.5}$, PM_{10}, SO_2, O_3, NO_2, and CO using historical pollutant measurements together with meteorological variables such as temperature, humidity, wind speed, precipitation, and atmospheric pressure [20].

Numerous comparative studies consistently report that LSTM models outperform traditional statistical approaches, support vector machines, decision trees, and conventional neural networks in forecasting accuracy. Their robustness under varying meteorological conditions has resulted in widespread adoption for both short-term and long-term air quality prediction [26].

Applied ANNs and LSTM models in forecasting some pollutants usually found in the AQI [27]. They found that the LSTM model achieved the best overall performance, recording the largest CC (0.87) and KTC (0.70) values, alongside the lowest MB (0.06), MFB (0.09), RMSE (0.14), and MAE (0.11) for all the pollutants. It also produced the smallest SD (0.01), reflecting superior precision and reliability in predictions.

Three different LSTM approaches, LSTM for regression with time steps, LSTM for regression using window and LSTM for regression, were applied by [28], to predict $PM_{2.5}$. The three variants of LSTM model all performed equally well in predicting $PM_{2.5}$ concentrations. The experimental findings showed that the R^2 consistently remained around 0.9 across all LSTM model variants. The combination of low RMSE values and high R^2 scores confirms the robustness and reliability of the model. The research in [29] improved the station-based Long Short-Term Memory (LSTM) model for spatial air quality forecasting by incorporating a novel deep-learning component, the broadcasting layer, which includes a learnable weight decay parameter to enable point-to-area prediction. Validation using 48-hour forecasts of $PM_{2.5}$ and O_3 demonstrated lower RMSE values, indicating superior performance compared to benchmark models.

Nevertheless, LSTM models generally require relatively large training datasets and considerable computational resources. Training complexity also increases substantially with longer input sequences, motivating the development of more computationally efficient recurrent architectures.

3.4 Hybrid Deep Learning Models

Recent research has increasingly focused on hybrid deep learning architectures that combine complementary modelling capabilities within a unified framework. The underlying motivation is that no individual architecture adequately captures every aspect of the complex spatiotemporal behaviour of atmospheric pollution.

The authors [30], employs remote sensing (RS), geographic information systems (GIS), and a hybrid LSTM framework to forecast $PM_{2.5}$ levels. The developed OA-LSTM model demonstrated higher performance, achieving a minimum RMSE of 3.01 $\mu g/m^3$ and a maximum R^2 of 0.88, highlighting its effectiveness relative to alternative models. Dev-LSTM, presented by [31], is a hybrid deconvolution-LSTM model that joins deconvolution with LSTM and capable of precise air pollutant levels predictions in both temporal and spatial dimensions.

In [32] is a hybrid Binary Gray Wolf Optimizer–DNN (BGWO-DNN) approach for air pollution prediction, achieving an R^2 of 88.21% and 93.17% accuracy, outperforming other models. Similarly [33], developed a dual graph convolution LSTM network using infrared sensors for CO_2, temperature, and humidity, along with geostatistical methods and low-power wireless networking, achieving up to 95% accuracy, surpassing existing techniques.

The authors in [34] introduced **DeepAir**, a framework designed to estimate fine-scale $PM_{2.5}$ concentrations from sparsely distributed data. By integrating the LightGBM algorithm with a pre-trained convolutional neural network, the model achieves high accuracy and efficient performance in both training and prediction tasks.

A Particle Swarm Optimisation (PSO) algorithm is used to determined optimal structure for the model's operation. This novel predictor, termed the Smart Air Quality Prediction Model (SAQPM), integrates intelligent computation by combining unsupervised learning through Long LSTM networks with optimisation via PSO [35],. The main objective of SAQPM is to forecast the concentrations of the key air pollutants that is particulate matter fine and coarse particles, nitrogen dioxide, carbon monoxide, ozone and sulfur dioxide.

W-BiLSTM(PSO)-GRU and XGBoost are combined to create an integrated hybrid framework for air quality prediction [36]. The suggested model uses wavelet decomposition and PSO-based parameter optimisation to improve prediction resilience, accuracy, and stability. The results reveal that the R^2 values for $PM_{2.5}$, PM_{10}, SO_2, CO, NO_2, and O_3 predictions were above 0.94, and the MAE and RMSE were below 0.02 and 0.03, respectively.

Similarly, the work in [37] combines a gated recurrent unit (GRU), a graph convolutional neural network (GCN), and a self-tuning spatio-temporal neural network (ST2NN) for improved air quality prediction. With an accuracy of 0.51%-10.18% measured using R2, the suggested approach outperformed other cutting-edge models.

These hybrid approaches consistently achieve superior predictive accuracy compared with standalone architectures because they simultaneously exploit spatial correlations, temporal dependencies, and multiscale environmental information. Hybrid models have therefore become the dominant trend in contemporary air pollution prediction research.

Despite their excellent predictive capability, hybrid models introduce increased architectural complexity, longer training times, greater computational cost, and reduced interpretability, presenting important challenges for operational deployment.

3.5 Transformer and Attention-Based Models

The introduction of attention mechanisms and Transformer architectures has significantly advanced deep learning for environmental prediction. Unlike recurrent architectures that process sequential observations one-time step at a time, Transformers analyse complete sequences simultaneously using self-attention mechanisms, enabling more effective modelling of long-range dependencies [38, 39].

Attention mechanisms further enhance predictive performance by automatically identifying the most informative temporal or spatial features within complex environmental datasets. This capability improves both forecasting accuracy and model interpretability [40].

Recent studies demonstrate that Transformer-based models consistently outperform conventional recurrent architectures for long-term forecasting, particularly when trained on large multisource datasets containing meteorological variables, satellite observations, traffic information, and pollutant measurements [41, 42].

Research conducted in [43] shows that transformer models outperform other deep learning models for air quality prediction. The proposed work is a lightweight deep learning model that learns complex relationships and long-term dependencies from time-series $PM_{2.5}$ data using sparse attention-based Transformer networks (STNs) with a multi-head sparse attention mechanism. With the Taizhou $PM_{2.5}$ datasets, the model performed exceptionally well, reaching an R^2 of 0.924 and reducing RMSE to 5.79 µg/m3 and MAE to 3.76 µg/m3.

However, these models generally require substantially greater computational resources and larger training datasets than recurrent neural networks, limiting their practical deployment in resource-constrained environments.

3.6 Transfer Learning and Emerging Architectures

Transfer learning has become increasingly important in environmental modelling because labelled air quality datasets remain limited in many developing countries [44]. By adapting knowledge learned from large benchmark datasets, transfer learning enables robust prediction even when local observations are scarce [45].

More recently, Graph Neural Networks (GNNs), Spatio-Temporal Graph Convolutional Networks (STGCNs), Physics-Informed Neural Networks (PINNs), and Foundation Models have emerged as promising research directions [46]. These models explicitly represent spatial relationships among monitoring stations, incorporate atmospheric physics into model learning, and facilitate large-scale environmental prediction across geographically diverse regions

[47, 48]. In Fig. 9 is a transfer learning architecture for air quality prediction.

For effective and precise air quality prediction, a two-phase attention model (TL-AdaBiGRU) based on transfer learning was presented in [49]. The model's first stage divides the air pollutant sequences into periods using a temporal distribution characterisation approach, while the second stage introduces a multi-head external attention mechanism to extract inherent features. The model's mean absolute error (MAE), root mean square error (RMSE), and mean absolute percentage error (MAPE) were reduced by 14%, 13%, and 4%, respectively, when compared to the previous models. Tables 3 and 4 are comparative analysis and representative studies of deep learning models for air pollution, forecasting, and prediction.

3.7 Graph Neural Networks (GNNs)

Graph Neural Networks (GNNs) have recently emerged as a promising deep learning paradigm for air pollution prediction because they effectively model the spatial relationships among air quality monitoring stations and other geographically distributed data sources [50, 51]. Unlike conventional deep learning models that assume regular grid-structured inputs, GNNs operate on graph-structured data where nodes represent monitoring stations or geographical locations and edges capture spatial correlations based on geographical proximity, atmospheric transport, or road networks [52]. This capability enables GNNs to model complex spatial dependencies and pollutant dispersion processes more effectively than traditional CNNs or recurrent models [53]. Recent studies have demonstrated that Graph Convolutional Networks (GCNs), Graph Attention Networks (GATs), and Spatio-Temporal Graph Neural Networks (STGNNs) achieve superior prediction accuracy by jointly learning spatial interactions and temporal dynamics [54]. Despite these advantages, GNNs require careful graph construction, substantial computational resources, and high-quality spatial data, which remain challenges for large-scale operational deployment [55].

3.8 Autoencoders

Autoencoders (AEs) are unsupervised deep learning models designed to learn compact and informative representations of high-dimensional data by encoding input features into a lower-dimensional latent space and subsequently reconstructing the original input. In air pollution prediction,

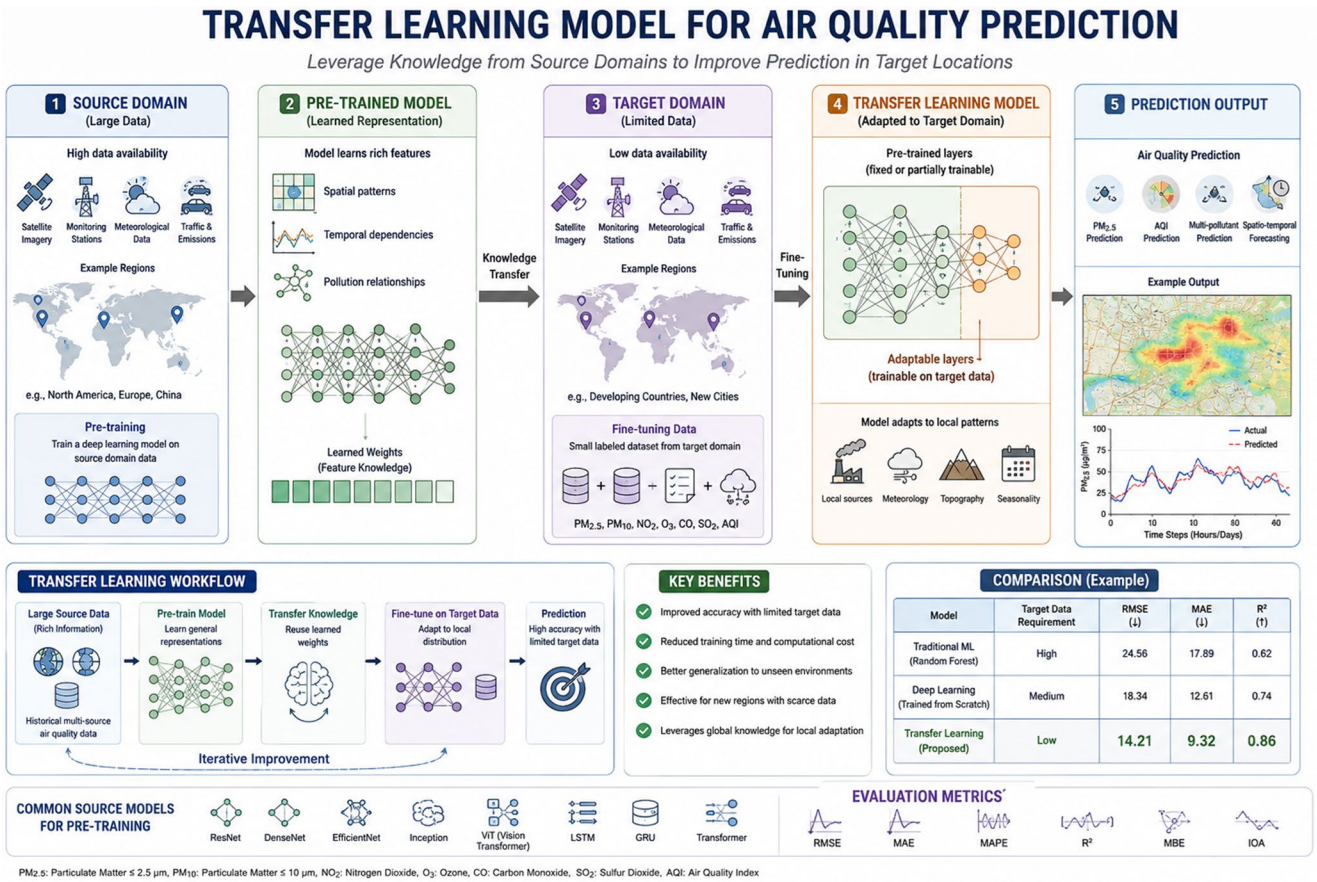


Fig. 9 Transfer learning framework for air quality prediction

Table 3 Comparative analysis studies

Deep Learning Model	Architecture Characteristics	Strengths	Limitations	Typical Applications	Common Input Data	Model Complexity
Artificial Neural Network (ANN)	Multi-layer feedforward neural network with fully connected neurons.	Simple implementation; captures nonlinear relationships; suitable for small datasets.	Poor scalability; limited ability to model spatial and temporal dependencies; prone to overfitting.	AQI prediction, pollutant concentration estimation, environmental regression.	Historical pollutant concentrations, meteorological variables.	Low–Moderate
Convolutional Neural Network (CNN)	Convolution and pooling layers extract hierarchical spatial features from structured data.	Excellent spatial feature extraction; automatic feature learning; robust to noise.	Limited capability for long-term temporal modelling; computationally intensive for deep architectures.	Pollution source identification, satellite image analysis, spatial $PM_{2.5}$ prediction, feature extraction.	Satellite imagery, air quality maps, GIS, gridded environmental data.	Moderate–High
Recurrent Neural Network (RNN)	Recurrent connections capture sequential dependencies in time-series data.	Models temporal dynamics effectively; suitable for sequential observations.	Vanishing/exploding gradient problems; weak long-term memory.	Short-term pollutant prediction, AQI forecasting.	Historical pollutant concentrations, meteorological time series.	Moderate
Long Short-Term Memory (LSTM)	Memory cells with input, forget, and output gates enable long-term dependency learning.	Excellent long-term temporal prediction; robust performance for multivariate time series.	Longer training time; large computational and memory requirements.	$PM_{2.5}$ prediction, AQI forecasting, multistep air quality prediction.	Historical pollutants, weather variables, traffic data.	High
Gated Recurrent Unit (GRU)	Simplified recurrent architecture using update and reset gates.	Faster training than LSTM; fewer parameters; competitive predictive performance.	Slightly less expressive for highly complex long-term dependencies.	Air pollution prediction, multivariate time-series forecasting.	Pollutant concentrations, meteorological observations.	Moderate–High
Transformer	Self-attention mechanism captures global temporal relationships without recurrent connections.	Superior long-range dependency modelling; highly parallelizable; state-of-the-art prediction accuracy.	Requires large datasets and high computational resources; reduced interpretability.	Long-term AQI prediction, $PM_{2.5}$ forecasting, spatiotemporal modelling.	Multivariate environmental time series, remote sensing, meteorological data.	Very High
Graph Neural Network (GNN)	Graph-based architecture models relationships among interconnected spatial locations using nodes and edges.	Captures complex spatial dependencies; suitable for irregular monitoring networks; excellent spatial reasoning.	Graph construction can be complex; computationally expensive for large-scale networks; requires high-quality spatial information.	Urban air quality prediction, sensor network modelling, regional pollution forecasting.	Monitoring station networks, road networks, GIS, satellite and meteorological data.	High
Autoencoder (AE)	Unsupervised encoder–decoder architecture learns compact latent feature representations.	Effective for feature extraction, dimensionality reduction, denoising, and missing data imputation.	Primarily used for representation learning rather than direct prediction; reconstruction quality depends on network design.	Data preprocessing, feature learning, anomaly detection, sensor calibration.	Multisource environmental datasets, sensor observations, meteorological variables.	Moderate
Transfer Learning (TL)	Reuses knowledge from pretrained models and adapts it to target prediction tasks.	Reduces training time; improves performance with limited labelled data; enhances model generalisation.	Performance depends on similarity between source and target domains; risk of negative transfer.	Air quality prediction in data-scarce regions, remote sensing applications.	Satellite imagery, environmental datasets, pretrained deep models.	Moderate
Federated Learning (FL)	Distributed collaborative learning where models are trained locally and aggregated without sharing raw data.	Preserves data privacy; enables collaborative model development; supports distributed environmental monitoring.	Communication overhead; heterogeneous local datasets; slower convergence; synchronization challenges.	Cross-city air pollution prediction, IoT sensor networks, smart city monitoring.	Distributed sensor networks, IoT devices, environmental agency databases.	High
Hybrid Deep Learning Models	Integrate two or more deep learning architectures (e.g., CNN–LSTM, CNN–GRU, Transformer–LSTM, GNN–LSTM).	Exploit complementary spatial and temporal features; consistently achieve superior prediction accuracy and robustness.	Increased model complexity; longer training time; higher computational cost; reduced interpretability.	Spatiotemporal air quality prediction, multistep forecasting, intelligent environmental monitoring.	Multi-source environmental datasets combining pollutants, weather, satellite imagery, traffic, and GIS.	Very High

Table 4 Representative studies

Model Family	Representative Studies	Primary Objective	Pollutant	Data Sources	Key Findings
Artificial Neural Networks (ANN)	Early studies (pre-2018)	Air quality prediction	$PM_{2.5}$, PM_{10}, NO_2	Ground monitoring stations, meteorological data	Demonstrated the feasibility of nonlinear pollutant prediction but showed limited capability for modelling complex spatiotemporal relationships.
Convolutional Neural Networks (CNN)	Recent CNN-based studies	Pollution, source localisation, spatial mapping	$PM_{2.5}$, PM_{10}, NO_2	Satellite imagery, remote sensing, GIS, land-use data	CNNs consistently achieved superior spatial feature extraction, enabling accurate pollution hotspot and high-resolution pollution mapping.
Transfer Learning (CNN-based)	Transfer learning approaches	Image-based pollution	$PM_{2.5}$	Satellite imagery, limited labelled datasets	Pre-trained CNN models significantly improved prediction accuracy and reduced training time, particularly for regions with limited data availability.
Recurrent Neural Networks (RNN)	Early sequential prediction studies	Time-series forecasting	$PM_{2.5}$, AQI	Historical air quality observations	Demonstrated the importance of temporal modelling but suffered from vanishing gradient problems that limited long-term prediction accuracy.
Long Short-Term Memory (LSTM)	Numerous forecasting studies	Short- and long-term air quality prediction	$PM_{2.5}$, PM_{10}, NO_2, SO_2, CO, O_3	Ground sensors, meteorological data, IoT sensors	LSTM remains the most widely adopted architecture for air pollution forecasting because of its ability to learn long-term temporal dependencies and consistently outperform traditional machine learning methods.
Bidirectional LSTM (BiLSTM)	Bidirectional forecasting studies	Multistep prediction	$PM_{2.5}$, AQI	Environmental time-series datasets	BiLSTM improved forecasting accuracy by simultaneously learning forward and backward temporal dependencies.
Gated Recurrent Unit (GRU)	GRU-based forecasting studies	Air quality prediction	$PM_{2.5}$, PM_{10}	Sensor networks, weather observations	GRU achieved performance comparable to LSTM while requiring fewer trainable parameters and shorter training times, making it suitable for real-time applications.
Attention-Based Models	Attention-enhanced LSTM and GRU studies	Long-term forecasting	$PM_{2.5}$, AQI	Multi-source environmental datasets	Attention mechanisms improved prediction accuracy by automatically identifying the most informative spatial and temporal features.
Transformer Models	Transformer and Informer studies	Long-term forecasting	$PM_{2.5}$, PM_{10}, AQI	Satellite imagery, meteorological observations, sensor networks	Transformer architectures demonstrated state-of-the-art performance for long-range prediction by effectively modelling global temporal dependencies using self-attention.
CNN–LSTM Hybrid Models	CNN–LSTM studies	Joint and forecasting	$PM_{2.5}$, PM_{10}	Satellite imagery, weather data, monitoring stations	CNN–LSTM architectures consistently outperformed standalone CNN and LSTM models by simultaneously capturing spatial and temporal information.
CNN–GRU Hybrid Models	CNN–GRU studies	Air quality prediction	$PM_{2.5}$	Sensor networks, meteorological data	CNN–GRU models achieved high prediction accuracy with lower computational cost than CNN–LSTM, making them attractive for real-time monitoring systems.
Transformer–LSTM Hybrid Models	Recent hybrid architectures	Large-scale forecasting	$PM_{2.5}$, AQI	Multi-source environmental datasets	Combining Transformer and LSTM architectures improved robustness, long-term prediction accuracy, and adaptability across heterogeneous datasets.

autoencoders are primarily employed for feature extraction, dimensionality reduction, missing data imputation, anomaly detection, and noise removal before predictive modelling [56]. Variants such as Stacked Autoencoders (SAEs), Sparse Autoencoders, Denoising Autoencoders (DAEs), and Variational Autoencoders (VAEs) have demonstrated the ability to improve data quality and enhance the predictive performance of downstream deep learning models. Autoencoders are particularly valuable when integrating heterogeneous environmental datasets, including meteorological variables, satellite imagery, and sensor observations, by extracting robust latent representations [57]. However, their effectiveness depends on careful architecture design and sufficient training data, and they are generally used as feature learning components rather than standalone prediction models [58].

3.9 Federated Learning

Federated Learning (FL) is an emerging distributed machine learning paradigm that enables multiple organisations or sensing networks to collaboratively train deep learning models without exchanging raw data. Instead of centralising environmental datasets, local models are trained independently at participating sites, and only model parameters or gradients are shared with a central aggregation server. This approach preserves data privacy while facilitating

collaborative model development across geographically distributed monitoring networks [59]. In air pollution prediction, Federated Learning has gained increasing attention for integrating data collected from smart cities, IoT sensor networks, and environmental agencies while addressing privacy, security, and data governance concerns. Furthermore, FL supports the development of more robust and generalisable prediction models by leveraging diverse environmental conditions from multiple regions [60]. Nevertheless, challenges such as communication overhead, heterogeneous data distributions, model convergence, and client synchronization continue to limit its widespread implementation [61]. Future research is expected to combine Federated Learning with Graph Neural Networks, Transformer architectures, and Edge AI to enable secure, scalable, and real-time intelligent air quality prediction systems.

Although still relatively new, these emerging architectures are expected to play an increasingly important role in next-generation intelligent air quality monitoring systems.

3.10 Summary of Deep Learning Model Evolution

The evolution of deep learning models demonstrates a clear progression from shallow neural networks towards increasingly sophisticated architectures capable of learning complex spatial and temporal relationships. CNNs remain the preferred choice for spatial pollution prediction and remote sensing applications, whereas LSTM and GRU networks continue to dominate temporal forecasting tasks. Attention mechanisms, Transformer models, and hybrid architectures currently represent the state of the art, consistently achieving the highest predictive accuracy across diverse environmental datasets. However, balancing predictive performance with computational efficiency, interpretability, and real-time deployment remains an important challenge that continues to drive research in this field.

4 Environmental Data Sources for Deep Learning-Based Air Pollution Monitoring (RQ2)

The predictive capability of a deep learning model depends not only on its architecture but also on the quality, diversity, and representativeness of the input data. Air pollution is a complex phenomenon influenced by meteorological conditions, emission sources, land-use characteristics, topography, and human activities. Consequently, recent studies increasingly employ heterogeneous data sources to improve model robustness and predictive accuracy [62–65].

The literature indicates a clear transition from using single-source monitoring data to integrating multiple complementary datasets. Traditional studies relied primarily on historical pollutant measurements obtained from fixed monitoring stations, whereas contemporary approaches combine ground observations with satellite imagery, meteorological variables, traffic information, geographic information systems (GIS), Internet of Things (IoT) sensor networks, and socioeconomic indicators. This multi-source data fusion enables deep learning models to capture both spatial and temporal characteristics of atmospheric pollution more effectively [30, 32, 66]. Figure 10 presents the taxonomy of environmental datasets commonly used in deep learning-based air pollution studies.

4.1 Ground-Based Air Quality Monitoring Stations

Ground-based monitoring stations remain the primary source of air quality observations and provide the reference measurements used in both model development and validation. These stations continuously measure pollutant concentrations using regulatory-grade instruments that comply with national and international environmental standards. The most frequently monitored pollutants include particulate matter ($PM_{2.5}$ and PM_{10}), nitrogen dioxide (NO_2), sulfur dioxide (SO_2), carbon monoxide (CO), ozone (O_3), and, in some regions, ammonia (NH_3) and volatile organic compounds (VOCs). Measurements are typically collected at hourly or daily intervals, providing long-term time-series data for forecasting models.

Because of their high measurement accuracy, ground monitoring stations serve as benchmark datasets for evaluating deep learning algorithms. LSTM and GRU models are particularly well suited to these sequential observations, while hybrid CNN–LSTM models often combine station data with spatial information from satellite imagery [23, 27, 67].

Despite their reliability, monitoring stations have several limitations. Their installation and maintenance costs restrict spatial coverage, especially in low- and middle-income countries. As a result, many regions lack sufficiently dense monitoring networks, leading to incomplete spatial representation of pollutant distributions.

4.2 Satellite Remote Sensing Data

Satellite observations have become one of the most valuable data sources for deep learning-based air pollution modelling because they provide continuous, large-scale spatial coverage that complements sparse ground monitoring networks [68, 69]. Remote sensing platforms such as MODIS, Sentinel, Landsat, VIIRS, and geostationary satellites provide measurements of aerosol optical depth (AOD), land surface temperature, vegetation indices, cloud properties, and

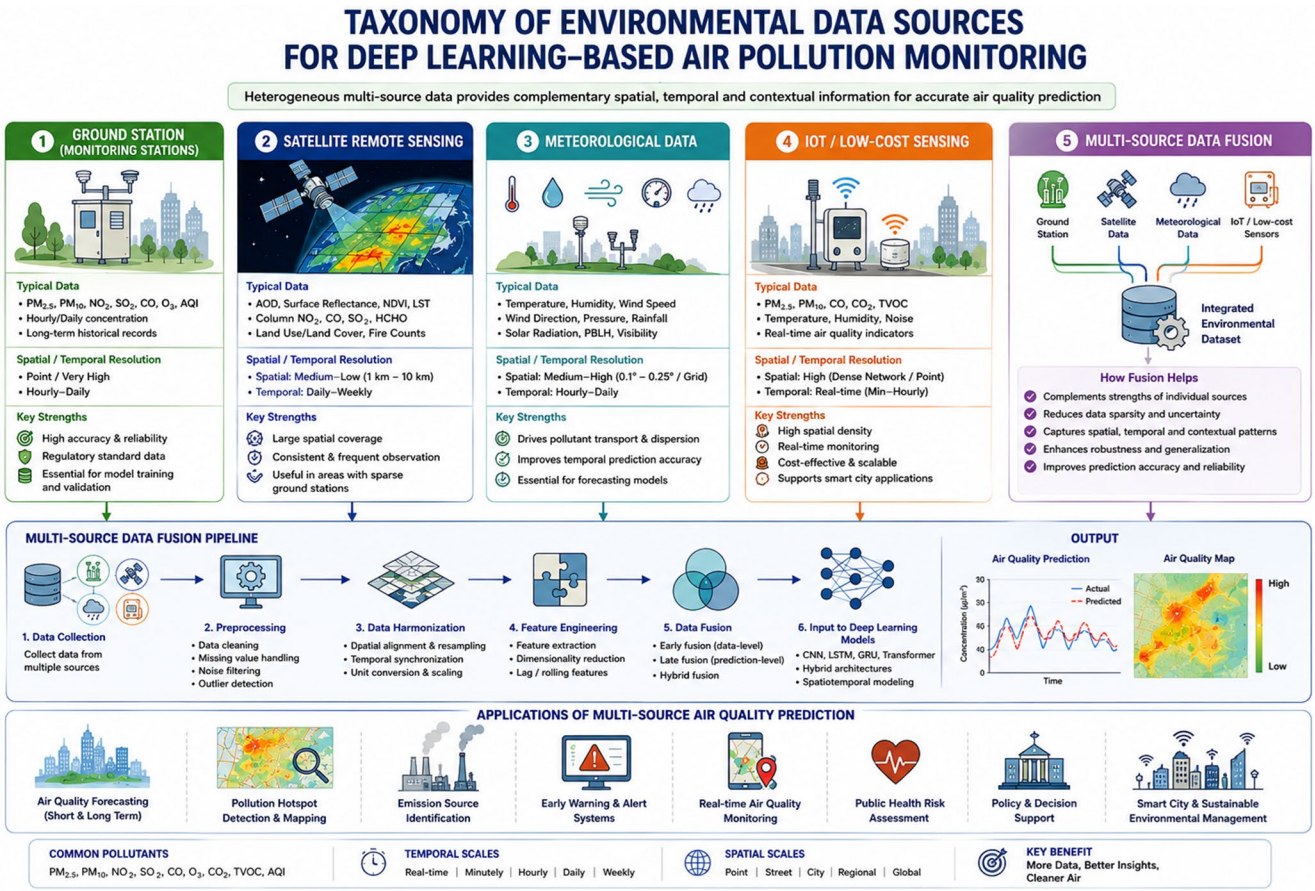


Fig. 10 Environmental datasets commonly used in deep learning-based air pollution studies

atmospheric composition. These datasets are widely used for estimating particulate matter concentrations, identifying pollution hotspots, and generating regional air quality maps.

CNN-based architectures dominate satellite image analysis because of their superior capability for extracting hierarchical spatial features. More recently, Transformer-based vision models have demonstrated promising performance in analysing multispectral remote sensing imagery [70, 71]. Nevertheless, satellite observations are indirect measurements of atmospheric pollutants and are susceptible to cloud cover, retrieval uncertainties, and variations in atmospheric conditions. Consequently, most recent studies combine satellite observations with ground monitoring measurements through data fusion techniques to improve prediction accuracy.

4.3 Meteorological Data

Meteorological variables play a critical role in determining the transport, dispersion, and accumulation of atmospheric pollutants. Consequently, virtually all modern deep learning models incorporate weather observations alongside pollutant concentrations. Frequently used meteorological variables include: Air temperature, Relative humidity, Wind speed, Wind direction, Atmospheric pressure, Rainfall, Solar radiation and Boundary layer height.

These variables influence pollutant transport, chemical reactions, and atmospheric mixing, enabling deep learning models to capture complex nonlinear relationships that cannot be represented using pollutant measurements alone. Numerous comparative studies demonstrate that incorporating meteorological variables substantially reduces forecasting errors and improves the generalisation capability of LSTM, GRU, and Transformer models [14, 72, 73].

4.4 Low-Cost Sensors and Internet of Things (IoT)

The rapid growth of Internet of Things technologies has transformed air quality monitoring by enabling the deployment of dense networks of low-cost sensors capable of collecting real-time environmental data. Compared with conventional monitoring stations, IoT sensors offer several advantages, including lower installation costs, greater spatial coverage, portability, and continuous monitoring. These characteristics make them particularly attractive for smart city applications and community-based environmental

monitoring [74, 75]. Recent deep learning studies increasingly utilise IoT datasets for real-time air quality forecasting, anomaly, and early warning systems. GRU and lightweight hybrid models are especially suitable for deployment on edge devices because of their relatively low computational requirements [76–78].

However, low-cost sensors generally produce noisier observations than regulatory-grade instruments and require calibration to compensate for environmental factors such as humidity and temperature. Consequently, preprocessing techniques including data cleaning, sensor calibration, and noise reduction remain essential components of deep learning workflows.

4.5 Multi-Source Data Fusion

A significant trend identified in the reviewed literature is the integration of multiple environmental datasets within unified deep learning frameworks [79, 80] Instead of relying on a single information source, contemporary studies combine:

- Ground monitoring station observations.
- Satellite remote sensing products.
- Meteorological variables.
- GIS and land-use information.
- Traffic and transportation data.
- IoT sensor measurements.
- Population density and socioeconomic indicators.

Multi-source data fusion enables models to capture complementary spatial and temporal information, leading to substantial improvements in forecasting accuracy and robustness. CNN–LSTM, Transformer-based, and attention-enhanced architectures are particularly effective because they can simultaneously process heterogeneous datasets [81, 82].

Despite these advantages, integrating multiple datasets introduces additional challenges, including differences in spatial resolution, temporal frequency, missing observations, and data synchronization. Addressing these issues remains an active area of research. In Table 5, we compare the principal environmental data sources, available air quality datasets used in deep learning-based air pollution research whereas Table 6 shows publicly available air quality datasets, their advantages and their common applications.

The following key findings were observed from the reviewed literature and have been summarised below:

- Ground-based monitoring stations remain the benchmark data source because of their high accuracy and long-term observational records.
- Satellite remote sensing provides extensive spatial coverage and is particularly valuable in regions where ground monitoring infrastructure is sparse.
- Meteorological variables are incorporated into nearly all state-of-the-art forecasting models because weather conditions strongly influence pollutant formation, transport, and dispersion.
- IoT and low-cost sensor networks have become increasingly important for real-time urban air quality monitoring despite requiring calibration to address measurement uncertainties.
- Multi-source data fusion has emerged as the dominant approach in recent studies, enabling hybrid deep learning architectures to exploit complementary information from heterogeneous environmental datasets and consistently achieve superior predictive performance.

From the studies, it was observed that Ground monitoring datasets (e.g., EPA AQS, CNEMC, EEA) remain the primary benchmark datasets because of their high measurement accuracy and regulatory standards. Also, Satellite datasets (e.g., MODIS, Sentinel-5P, Landsat, Sentinel-2) provide extensive spatial coverage and are frequently combined with ground observations for pollution mapping and regional forecasting.

Meteorological reanalysis datasets (e.g., ERA5 and MERRA-2) on the other hand, are widely integrated with pollutant measurements to improve the prediction of pollutant transport and dispersion. Global open-access repositories like OpenAQ and WAQ were mostly used for transfer learning tasks, cross-regional model development, and benchmarking due to their extensive geographical coverage. It was also observed that combining ground observations, satellite imagery, meteorological variables, and IoT sensor data, has become the dominant strategy in recent deep learning studies because it consistently improves prediction accuracy, robustness, and model generalisation. In Table 7 is a summary of frequently used input variables in deep learning models for air pollution forecasting and prediction.

This study also revealed several consistent trends regarding feature selection which has been categorised under the following:

1. Historical pollutant concentrations (particularly $PM_{2.5}$ and PM_{10}) remain the most influential predictors because pollutant levels exhibit strong temporal autocorrelation.
2. Meteorological variables are incorporated into nearly all state-of-the-art forecasting models. Temperature, relative humidity, wind speed, and atmospheric pressure are the most frequently used environmental variables because they strongly influence pollutant transport, dispersion, and chemical transformation.

Table 5 Comparison of environmental data sources used in deep learning-based air pollution monitoring

Data Source	Spatial Resolution	Temporal Resolution	Advantages	Limitations	Common Applications
Ground-Based Monitoring Stations	Point locations (station-based)	Real-time, hourly, daily	High measurement accuracy; regulatory-grade observations; reliable benchmark for model training and validation; continuous long-term records	Sparse spatial coverage; high installation and maintenance costs; limited availability in developing countries	Air Quality Index (AQI) estimation, pollutant concentration prediction, model validation, time-series forecasting
Satellite Remote Sensing	Regional to global (10 m–10 km depending on sensor)	Daily to weekly (sensor-dependent)	Wide spatial coverage; suitable for regions lacking monitoring stations; supports pollution mapping and aerosol monitoring; freely available datasets for many missions	Cloud contamination; indirect pollutant estimation; lower temporal frequency; atmospheric retrieval uncertainties	$PM_{2.5}$ estimation, pollution hotspot, aerosol optical depth (AOD) analysis, land-use assessment, regional air quality mapping
Meteorological Data	Station-level to regional grids	Hourly to daily	Captures environmental drivers of pollutant dispersion; improves prediction accuracy; readily available from weather agencies and reanalysis products	Weather variables alone cannot represent pollution sources; uncertainty in numerical weather predictions	Air pollution forecasting, dispersion modelling, multivariate prediction, meteorology-aware deep learning models
Low-Cost Sensors / IoT Networks	Dense local or urban-scale coverage	Near real-time (seconds to minutes)	Low deployment cost; high spatial density; continuous monitoring; suitable for smart cities and edge computing	Lower accuracy than regulatory instruments; sensor drift; frequent calibration and maintenance required; susceptible to environmental interference	Real-time monitoring, anomaly, community air quality monitoring, early warning systems, smart city applications
Geographic Information Systems (GIS) and Land-Use Data	High-resolution spatial layers	Static or periodically updated	Provides spatial context; supports modelling of urban morphology, road networks, vegetation, and industrial zones; enhances spatial prediction	Static datasets may not reflect rapid environmental changes; requires integration with dynamic observations	Pollution source localisation, spatial feature extraction, land-use regression, urban air quality mapping
Traffic and Transportation Data	Road segment or city level	Minutes to hourly	Captures mobile emission sources; improves urban pollution prediction; reflects congestion dynamics	Data accessibility may be limited; highly variable across locations and time; influenced by traffic management policies	Urban pollution forecasting, roadside air quality prediction, emission modelling, smart transportation systems
Industrial Emission Inventories	Facility or regional level	Monthly, annual, or event-based	Identifies major stationary emission sources; useful for emission trend analysis; complements monitoring observations	Often updated infrequently; may contain reporting uncertainties; limited accessibility in some regions	Source apportionment, emission forecasting, industrial pollution assessment, policy evaluation
Population and Socio-economic Data	Administrative units or gridded datasets	Annual or census intervals	Captures human activity patterns; useful for exposure assessment and environmental justice studies; supports demand forecasting	Low temporal resolution; indirect relationship with pollutant concentrations; varying data quality	Population exposure analysis, health impact assessment, urban planning, environmental risk mapping
Multi-Source Data Fusion	Multi-scale (local to global)	Multi-temporal (real-time to annual)	Integrates complementary information; improves prediction accuracy and robustness; captures spatial and temporal dependencies simultaneously; supports advanced hybrid deep learning models	Complex preprocessing; data synchronisation challenges; increased computational cost; heterogeneous data formats	Hybrid deep learning (CNN–LSTM, CNN–GRU, Transformer), multimodal learning, intelligent air quality monitoring, operational forecasting systems

Table 6 Publicly available air quality datasets

Dataset	Provider	Pollutants / Variables	Geographical Coverage	Temporal Resolution	Common Applications
U.S. EPA Air Quality System (AQS)	United States Environmental Protection Agency (EPA)	$PM_{2.5}$, PM_{10}, NO_2, SO_2, CO, O_3, Pb, AQI	United States	Hourly, Daily	Air quality prediction, AQI forecasting, model benchmarking, time-series analysis
OpenAQ	OpenAQ Foundation	$PM_{2.5}$, PM_{10}, NO_2, SO_2, CO, O_3, BC and others	Global (160 + countries)	Real-time, Hourly	Global air quality monitoring, transfer learning, cross-country prediction, data fusion
World Air Quality Index (WAQI)	World Air Quality Index Project	AQI, $PM_{2.5}$, PM_{10}, NO_2, SO_2, CO, O_3	Global	Real-time, Hourly	AQI prediction, visualization, smart city monitoring, public health applications
China National Environmental Monitoring Centre (CNEMC)	Ministry of Ecology and Environment, China	$PM_{2.5}$, PM_{10}, NO_2, SO_2, CO, O_3	China	Hourly	Deep learning forecasting, $PM_{2.5}$ prediction, urban pollution modelling
European Environment Agency (EEA) Air Quality e-Reporting	European Environment Agency	$PM_{2.5}$, PM_{10}, NO_2, SO_2, CO, O_3, Ben ne	Europe	Hourly, Daily	Air quality forecasting, environmental policy evaluation, regional pollution studies
AirNow	U.S. EPA and partner agencies	AQI, $PM_{2.5}$, PM_{10}, O_3	United States, Canada, Mexico	Hourly	AQI forecasting, public health alerts, real-time prediction
Copernicus Atmosphere Monitoring Service (CAMS)	European Centre for Medium-Range Weather Forecasts (ECMWF)	$PM_{2.5}$, PM_{10}, O_3, NO_2, SO_2, CO, Aerosols	Global	Hourly, Daily	Air pollution forecasting, atmospheric chemistry modelling, data fusion
NASA MODIS (Terra/Aqua)	National Aeronautics and Space Administration (NASA)	Aerosol Optical Depth (AOD), Land Surface Temperature (LST), NDVI	Global	Daily	$PM_{2.5}$ estimation, remote sensing, spatial pollution mapping
Sentinel-5P (TROPOMI)	European Space Agency (ESA)	NO_2, SO_2, CO, O_3, CH_4, Aerosol Index	Global	Daily	Satellite-based pollutant monitoring, emission, regional forecasting
Landsat 8/9	United States Geological Survey (USGS) & NASA	Land Cover, Surface Temperature, Vegetation Indices	Global	16-day revisit	Land-use analysis, pollution source identification, spatial modelling
Sentinel-2	European Space Agency (ESA)	Land Cover, Vegetation Indices, Surface Reflectance	Global	5-day revisit	Urban environmental monitoring, land-use classification, CNN-based feature extraction
ERA5 Reanalysis	ECMWF	Temperature, Humidity, Wind Speed, Pressure, Rainfall, Solar Radiation	Global	Hourly	Meteorology-assisted forecasting, LSTM and Transformer models
MERRA-2	NASA Global Modeling and Assimilation Office	Aerosols, Meteorological Variables, Atmospheric Composition	Global	Hourly	Atmospheric modelling, $PM_{2.5}$ estimation, climate–air quality interaction studies
PurpleAir Network	PurpleAir	$PM_{1.0}$, $PM_{2.5}$, PM_{10}	Global (community sensor network)	Real-time (2–10 min)	IoT-based monitoring, low-cost sensor calibration, anomaly
UCI Machine Learning Repository – Beijing $PM_{2.5}$ Dataset	UCI Machine Learning Repository	$PM_{2.5}$, Temperature, Pressure, Wind Speed, Dew Point	Beijing, China	Hourly	Benchmark dataset for machine learning and deep learning forecasting
Kaggle Air Quality Datasets	Kaggle Community	Pollutants, Weather, AQI (varies by dataset)	Multiple countries	Hourly to Daily	Algorithm benchmarking, educational research, comparative modelling

Abbreviations: AQI - Air Quality Index; $PM_{2.5}$- Particulate Matter≤2.5 µm; PM_{10}- Particulate Matter≤10 µm; NO_2 - Nitrogen Dioxide; SO_2- Sulfur Dioxide; CO - Carbon Monoxide; O_3 - Ozone; Pb - Lead; AOD - Aerosol Optical Depth; LST - Land Surface Temperature; NDVI - Normalised Difference Vegetation Index

Table 7 Frequently used input variables

Variable Category	Common Variables	Purpose in Deep Learning Models	Frequently Used Models	Importance
Air Pollutants	$PM_{2.5}$, PM_{10}, NO_2, SO_2, CO, O_3, NH_3, VOCs, AQI	Primary prediction targets and historical inputs for forecasting pollutant concentrations	LSTM, GRU, Transformer, CNN–LSTM	Very High
Meteorological Variables	Temperature, Relative Humidity, Wind Speed, Wind Direction, Atmospheric Pressure, Rainfall, Solar Radiation, Boundary Layer Height	Capture pollutant transport, dispersion, chemical reactions, and atmospheric stability	LSTM, GRU, Transformer, CNN–LSTM	Very High
Satellite-Derived Variables	Aerosol Optical Depth (AOD), Land Surface Temperature (LST), NDVI, Surface Reflectance, Aerosol Index, Cloud Fraction	Provide large-scale spatial information for pollutant estimation and remote sensing applications	CNN, Vision Transformer (ViT), CNN–LSTM	High
Temporal Variables	Hour of Day, Day of Week, Month, Season, Holidays, Weekends	Capture daily, weekly, and seasonal pollution patterns	LSTM, GRU, Transformer	High
Remote Sensing Products	MODIS AOD, Sentinel-5P NO_2, Sentinel-2 Land Cover, Landsat Surface Temperature	Enhance spatial prediction where monitoring stations are sparse	CNN, Transformer, Hybrid Models	High
IoT Sensor Measurements	$PM_{2.5}$, $PM_{10,}$ Temperature, Humidity, CO_2, TVOCs, Sensor Status	Enable real-time monitoring and smart city applications	GRU, CNN–GRU, Lightweight Deep Networks	Moderate–High
Weather Forecast Variables	Forecast Temperature, Forecast Wind Speed, Forecast Rainfall, Forecast Humidity	Improve multi-step forecasting by incorporating future meteorological conditions	LSTM, Transformer, Hybrid Models	High
Derived Statistical Features	Mean, Standard Deviation, Maximum, Minimum, Trend, Variance	Improve feature representation and reduce noise	ANN, LSTM, GRU	Moderate
Multi-Source Fusion Features	Combined Ground Monitoring, Satellite, Weather, Traffic, GIS, and IoT Variables	Capture complementary spatial and temporal information for robust prediction	CNN–LSTM, CNN–GRU, Transformer, GNN	Very High

3. Satellite-derived variables, especially Aerosol Optical Depth (AOD), have become increasingly important for estimating particulate matter concentrations in regions with limited ground monitoring infrastructure.
4. Multi-source feature fusion has become the dominant strategy in recent studies. Integrating ground observations, meteorological variables, satellite imagery, GIS information, and IoT sensor data consistently produce more accurate and robust deep learning models than relying on a single data source.
5. Recent Transformer-based and hybrid architectures increasingly incorporate heterogeneous datasets because attention mechanisms are particularly effective at learning relationships among diverse feature types.

4.6 Challenges Associated with Environmental Datasets

Although environmental datasets continue to increase in both quantity and diversity, several challenges remain. Common issues identified in the reviewed literature include:

- Missing observations resulting from sensor failures.
- Limited monitoring infrastructure in developing countries.
- Class imbalance caused by relatively few extreme pollution events.
- Inconsistent temporal and spatial resolutions across datasets.
- Noise and calibration errors in low-cost sensors.
- Cloud contamination in satellite imagery.
- Limited availability of labelled datasets for supervised learning.
- Privacy and accessibility constraints affecting real-time environmental data sharing.

These challenges directly influence model accuracy, generalisation, and operational deployment, highlighting the importance of robust preprocessing, feature engineering, and data fusion techniques.

4.7 Summary of Environmental Data Sources

The reviewed literature demonstrates that environmental datasets have evolved from isolated monitoring station observations towards heterogeneous multi-source data ecosystems. Ground monitoring stations continue to provide the reference measurements for model training and validation, while satellite imagery contributes extensive spatial coverage. Meteorological variables enhance temporal prediction, IoT sensors facilitate real-time monitoring. Increasingly, hybrid deep learning architectures integrate these complementary information sources through multimodal data fusion, producing more accurate and reliable air quality prediction systems. Nevertheless, issues related to data quality, availability, interoperability, and standardisation remain significant challenges that must be addressed to support next-generation intelligent air quality management.

5 Comparative Performance Analysis of Deep Learning Models for Air Pollution Forecasting and Prediction (RQ3)

Evaluating the performance of deep learning models is essential for determining their suitability for operational air quality monitoring systems. While numerous architectures have been proposed over the past decade, their effectiveness varies considerably depending on the prediction task, input variables, pollutant type, forecasting horizon, geographical location, and dataset characteristics [83, 84]. Consequently, direct comparison of model performance has become an important area of investigation. The reviewed literature indicates a clear evolution from conventional statistical methods towards increasingly sophisticated deep learning architectures capable of modelling complex nonlinear relationships and spatiotemporal dependencies. Although no single model consistently outperforms all others under every circumstance, recent this study generally reports superior performance for hybrid and attention-based architectures compared with standalone deep learning models. Figure 11 presents the framework used to evaluate deep learning models in air pollution applications.

5.1 Performance Evaluation Metrics

Performance evaluation metrics provide objective measures for assessing prediction accuracy and comparing competing deep learning architectures. The choice of evaluation metric largely depends on whether the task involves regression or classification. Regression metrics dominate the reviewed literature because most studies focus on predicting continuous pollutant concentrations. The most frequently reported metrics include: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), Coefficient of Determination (R^2) and Mean Squared Error (MSE) for regression tasks.

For pollution and classification tasks, researchers commonly employ: Accuracy, Precision, Recall (Sensitivity), Specificity, F1-score and Area Under the Receiver Operating Characteristic Curve (AUC) according to the reviewed literature. Table 8 summarises the performance metrics commonly used in deep learning-based air pollution, forecasting, and prediction.

5.2 Factors Influencing Model Performance

This section highlights on key factors identified in literature that affects model performance. This review identified

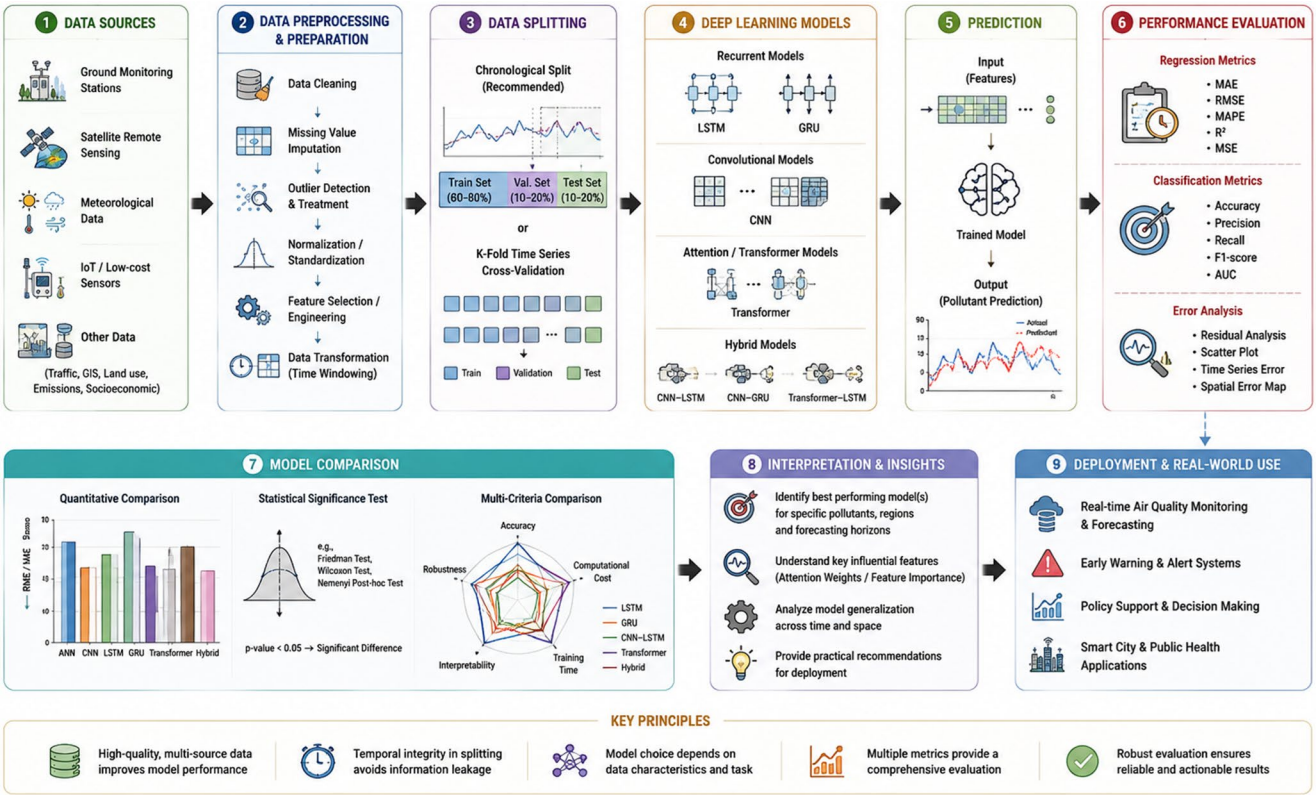


Fig. 11 Framework used to evaluate deep learning models in air pollution applications

several factors that significantly influence prediction accuracy and have been highlighted below:

5.2.1 Data Quality

Missing observations, sensor calibration errors, inconsistent sampling frequencies, and noisy measurements reduce model reliability.

Studies consistently report improved performance following preprocessing procedures such as interpolation, outlier removal, normalisation, and missing-value imputation.

Dataset Size: Large datasets generally improve model generalisation. Transformer architectures, in particular, benefit substantially from large-scale training datasets, whereas LSTM and GRU models remain effective with relatively moderate data volumes.

5.2.2 Feature Selection

Models trained using only pollutant concentrations generally produce lower prediction accuracy than those incorporating meteorological variables, satellite observations, and land-use information. Feature engineering and multimodal data fusion play important roles in improving predictive performance.

5.2.3 Forecasting Horizon

Short-term prediction (1–24 h) generally achieves considerably lower prediction errors than long-term forecasting extending several days into the future. Prediction uncertainty increases with forecasting horizon regardless of model architecture.

5.2.4 Regional Characteristics

Model performance varies according to geographical location because pollutant formation mechanisms, meteorological conditions, industrial activities, and transportation systems differ substantially across regions. Consequently, models developed in one region often require retraining before deployment elsewhere.

6 Research Gaps, Challenges, and Future Directions (RQ4)

The rapid advancement of deep learning has significantly improved the ability to detect, forecast, and predict air pollution. Despite these achievements, several technical, methodological, and practical challenges continue to limit the widespread deployment of intelligent air quality monitoring systems. The reviewed literature indicates that while prediction accuracy has improved considerably, issues related to data quality, model interpretability, computational efficiency, scalability, and operational deployment remain unresolved. Addressing these challenges is essential for developing reliable, transparent, and deployable air pollution prediction systems. Figure 12 summarises the major research gaps identified in the literature,

6.1 Data Availability and Quality Challenges

The effectiveness of deep learning models depends heavily on the availability of large, high-quality datasets. However, many studies continue to face significant challenges related to incomplete observations, missing values, noisy measurements, inconsistent sampling frequencies, and heterogeneous data formats.

Ground monitoring stations remain unevenly distributed across the world, with particularly limited coverage in low- and middle-income countries. Consequently, many developing regions lack sufficient historical observations for training robust deep learning models. Satellite observations partially alleviate this problem but are affected by cloud cover, atmospheric retrieval uncertainties, and relatively coarse temporal resolution.

Future research should focus on developing robust data imputation techniques, uncertainty-aware learning methods, and standardised data integration frameworks capable of effectively combining heterogeneous environmental datasets.

6.2 Limited Generalisation Across Regions

Most existing deep learning models are developed and validated using data collected from specific cities or countries. Consequently, models trained in one geographical region often exhibit reduced performance when applied elsewhere due to differences in climate, emission sources, urban morphology, and meteorological conditions. Improving cross-regional generalisation remains a major research challenge. Transfer learning, domain adaptation, and meta-learning techniques offer promising solutions by enabling knowledge acquired in one region to be transferred to locations with limited monitoring infrastructure.

Future studies should evaluate model robustness across multiple climatic zones and investigate globally transferable prediction frameworks.

6.3 Model Interpretability and Explainability

Although deep learning models generally achieve higher predictive accuracy than conventional machine learning

Table 8 Performance evaluation metrics commonly used in literature

Metric	Formula*	Category	Interpretation	Strengths	Limitations	Common Applications
Mean Absolute Error (MAE)	$MAE = \frac{1}{n}\sum_{\{i=1\}}^{n} \left\lvert \left(y_i - \widehat{y_i}\right) \right\rvert$	Regression	Averaged magnitude of errors	Easy to interpret;	less sensitive to outliers	
Mean Squared Error (MSE)	$MSE = \frac{1}{n}\sum_{\{i=1\}}^{n} \left(y_i - \widehat{y_i}\right)^2$	Regression	Average squared prediction error	Penalises large errors; useful during model optimization	Units are squared, making interpretation less intuitive	Model training, hyperparameter optimization
Root Mean Square Error (RMSE)	$RMSE = \sqrt{MSE}$	Regression	Square root of the average squared error	Widely used; emphasises large prediction errors; same units as the target variable	Sensitive to extreme outliers	Air quality forecasting, pollutant prediction, model benchmarking
Mean Absolute Percentage Error (MAPE)	$MAPE = \frac{100}{n}\sum_{\{i=1\}}^{n} \left\lvert \left(\frac{y_i - \widehat{y_i}}{y_i}\right) \right.$	Regression	Average percentage prediction error	Scale-independent; facilitates comparison across datasets	It fails with zero actuals (division by zero).	Produce intuitive output. Highly interpretable for non-technical stakeholders
Coefficient of Determination (R^2)	$R^2 = \frac{SSRes}{SSTot}$	Regression	Proportion of variance explained by the model	Indicates goodness of fit; easy to compare models	Does not directly quantify prediction error	Regression model evaluation, forecasting studies
Accuracy	$Acc = \frac{TP+TN}{TP+TN+FP+FN}$	Classification	Proportion of correctly classified observations	Simple and intuitive	Can be misleading for imbalanced datasets	Pollution event, AQI category classification
Precision	$Precision = \frac{TP}{TP+FP}$	Classification	Percentage of predicted positive cases that are correct	Measures false alarm rate	Ignores false negatives	Pollution episode, emission source identification
Recall	$Recall = \frac{TP}{TP+FN}$	Classification	Ability to correctly identify actual positive cases	Detects missed pollution events	May increase false positives	Air pollution event, anomaly
Specificity	$Specificity = \frac{TN}{TN+FP}$	Classification	Ability to correctly identify negative cases	Complements recall in binary classification	Less informative when negative cases dominate	Air quality classification
F1-Score	$F1 = 2 \times \frac{Precision \times Recall}{Precision+Recall}$	Classification	Harmonic mean of precision and recall	Balances false positives and false negatives	Does not consider true negatives	Imbalanced pollution classification tasks
Area Under the ROC Curve (AUC)	Area under the ROC curve	Classification	Ability of a model to distinguish between classes	Threshold-independent; robust classifier evaluation	Less informative when probabilities are poorly calibrated	Binary pollution, health risk classification



Fig. 12 Research gaps identified in literature

approaches, they are frequently criticised for functioning as "black-box" systems. Environmental scientists and policymakers increasingly require transparent models capable of explaining why particular predictions are generated. This is especially important when deep learning models are used to support public health decisions and environmental policy.

Recent studies have begun integrating Explainable Artificial Intelligence (XAI) techniques such as SHAP, LIME, Grad-CAM, and attention visualisation to improve model interpretability. However, explainability remains relatively underexplored within air pollution research [85, 86].

Future work should prioritise the development of inherently interpretable deep learning architectures that provide reliable explanations alongside accurate predictions.

6.4 Computational Complexity

Recent deep learning architectures, particularly Transformer-based and hybrid models, require substantial computational resources for training and deployment.

Large model sizes, long training times, and high energy consumption limit their applicability in resource-constrained environments such as embedded systems, mobile devices, and developing countries.

Future research should investigate lightweight deep learning architectures, model compression, pruning, quantisation, knowledge distillation, and efficient neural architecture search techniques that maintain high predictive accuracy while reducing computational requirements.

6.5 Real-time Deployment Challenges

Although numerous deep learning models demonstrate excellent predictive performance under experimental conditions, relatively few have been deployed in operational air quality monitoring systems. Real-time deployment introduces additional challenges including: Continuous data acquisition, sensor reliability, communication latency, edge computing constraints, model updating, system scalability, cybersecurity and energy efficiency.

Emerging Edge AI technologies provide promising opportunities for deploying lightweight deep learning models directly on IoT devices, enabling continuous local prediction with reduced communication overhead.

6.6 Multi-source Data Fusion Challenges

The literature consistently demonstrates that integrating multiple environmental datasets improves prediction accuracy. Nevertheless, effective multimodal learning remains challenging because environmental datasets often differ in spatial resolution, temporal frequency, measurement uncertainty, and data quality.

Future research should investigate advanced multimodal learning frameworks capable of automatically aligning heterogeneous environmental datasets while effectively handling missing information and uncertainty.

Transformer-based multimodal fusion architectures, graph neural networks, and foundation models appear particularly promising for addressing these challenges.

6.7 Benchmarking and Standardisation

A major limitation identified throughout the reviewed literature is the absence of standard benchmarking protocols. Many studies employ different datasets, forecasting horizons, evaluation metrics, and experimental settings, making direct comparison of model performance difficult.

Future research should encourage: Standard benchmark datasets, public evaluation platforms, common preprocessing protocols, reproducible experimental designs, open-source. implementations.

Improved benchmarking would facilitate fair comparison among competing deep learning architectures and accelerate scientific progress. Below is Table 9 that synthesises the major research gaps identified in the literature.

6.8 Emerging Research Directions

The literature indicates several rapidly emerging research themes that are expected to shape the next generation of intelligent air quality monitoring systems. In Fig. 13 is a pictorial representation of emerging research directions for next-generation deep learning-based air quality monitoring systems.

6.8.1 Explainable Artificial Intelligence (XAI)

Integrating explainability into deep learning models will increase user trust and facilitate adoption by environmental agencies and policymakers.

6.8.2 Graph Neural Networks (GNNs)

Graph Neural Networks effectively capture spatial relationships among monitoring stations, transportation networks, and urban infrastructure, making them highly suitable for large-scale urban air quality prediction.

6.8.3 Federated Learning

Federated learning enables collaborative model training across multiple organisations without sharing sensitive environmental data, thereby addressing privacy and data governance concerns.

6.8.4 Self-Supervised and Semi-Supervised Learning

Obtaining labelled environmental datasets remains expensive and time-consuming. Self-supervised and semi-supervised learning approaches provide promising alternatives by exploiting large quantities of unlabeled environmental data.

6.8.5 Edge Artificial Intelligence

Edge AI enables deep learning inference directly on low-power IoT devices, reducing latency, communication costs, and dependence on cloud infrastructure for real-time air quality monitoring.

6.8.6 Digital Twins and Smart Cities

Digital twin technologies integrate environmental sensors, remote sensing platforms, simulation models, and deep learning algorithms to create virtual representations of urban environments. These systems support real-time decision-making, scenario analysis, and adaptive pollution control strategies within smart cities.

6.9 Future Research Agenda

Based on the reviewed literature, several priority areas are recommended for future investigation and the same have been listed below for further exploration by scholars and researchers:

1. Develop globally transferable deep learning models that generalise across diverse climatic and geographical conditions.
2. Design lightweight architectures suitable for real-time deployment on edge and IoT devices.
3. Integrate Explainable Artificial Intelligence to improve transparency, interpretability, and user trust.
4. Expand multimodal data fusion by combining ground monitoring stations, satellite imagery, meteorological observations, traffic information, GIS, and socioeconomic variables.

Table 9 Summary of major research gaps identified in the literature

Research Gap	Description	Impact on Air Pollution Monitoring	Proposed Solutions	Research Priority
Data Scarcity and Limited Monitoring Infrastructure	Many regions, particularly low- and middle-income countries, lack sufficient ground-based monitoring stations and long-term air quality records.	Limits model training, reduces prediction accuracy, and hinders generalisation across regions.	Expand monitoring networks, integrate low-cost sensors with satellite observations, develop transfer learning and data augmentation techniques.	Very High
Poor Data Quality and Missing Values	Environmental datasets frequently contain missing observations, noise, calibration errors, and inconsistent sampling intervals.	Produces unreliable predictions and increases model uncertainty.	Develop robust preprocessing pipelines, advanced imputation methods, sensor calibration techniques, and uncertainty-aware learning algorithms.	Very High
Limited Generalisation Across Geographic Regions	Most models are trained and validated using data from a single city or country.	Reduces model transferability and operational deployment in diverse climatic conditions.	Employ domain adaptation, transfer learning, meta-learning, and cross-regional validation strategies.	High
Lack of Explainability and Interpretability	Most deep learning models operate as black boxes, making prediction decisions difficult to understand.	Reduces trust among policymakers, environmental agencies, and the public.	Integrate Explainable Artificial Intelligence (XAI) techniques such as SHAP, LIME, Grad-CAM, attention visualisation, and interpretable model architectures.	High
High Computational Cost	Transformer and hybrid deep learning models require significant computational resources, memory, and energy.	Limits deployment in resource-constrained environments and real-time applications.	Develop lightweight models through pruning, quantisation, knowledge distillation, neural architecture search, and efficient edge computing solutions.	High
Limited Real-Time Deployment	Most studies evaluate models under laboratory conditions rather than operational environments.	Restricts practical implementation for continuous air quality monitoring and early warning systems.	Design real-time IoT-enabled frameworks, Edge AI systems, adaptive model updating, and cloud–edge collaborative architectures.	High
Challenges in Multi-Source Data Fusion	Environmental datasets differ in spatial resolution, temporal frequency, formats, and uncertainty levels.	Reduces the effectiveness of multimodal learning and limits prediction robustness.	Develop advanced multimodal fusion techniques, attention mechanisms, graph neural networks, and data harmonisation frameworks.	High
Lack of Standard Benchmark Datasets	Studies use different datasets, preprocessing methods, forecasting horizons, and evaluation protocols.	Makes fair comparison of model performance difficult and reduces reproducibility.	Establish publicly available benchmark datasets, standard evaluation protocols, and open-source implementations.	Very High
Insufficient Research in Developing Countries	Most studies focus on China, the United States, and Europe, with limited representation from Africa and other developing regions.	Reduces understanding of pollution dynamics in underrepresented regions and limits global applicability.	Promote international collaboration, open-access datasets, and region-specific studies supported by local monitoring initiatives.	High
Limited Integration of Atmospheric Science	Deep learning models often ignore physical and chemical processes governing pollutant formation and transport.	Predictions may lack physical consistency and fail under changing environmental conditions.	Integrate physics-informed neural networks (PINNs), atmospheric chemistry models, and numerical weather prediction outputs.	High
Privacy and Data Sharing Constraints	Environmental data are often distributed across multiple organisations with legal and privacy restrictions.	Limits collaborative model development and large-scale data integration.	Adopt federated learning, secure multi-party computation, and privacy-preserving machine learning techniques.	Moderate–High
Limited Adoption of Emerging AI Technologies	Technologies such as foundation models, self-supervised learning, and large-scale multimodal AI remain underexplored in air pollution research.	Slows innovation and limits exploitation of recent advances in artificial intelligence.	Investigate foundation models, multimodal Transformers, self-supervised learning, and large language model-assisted environmental analytics.	Moderate–High
Model Robustness and Uncertainty Quantification	Most studies report prediction accuracy without assessing uncertainty or robustness under changing environmental conditions.	Reduces confidence in operational decision-making and risk assessment.	Incorporate Bayesian deep learning, probabilistic forecasting, ensemble learning, and uncertainty estimation techniques.	High
Energy Efficiency and Sustainability	Training deep learning models requires considerable computational energy and carbon emissions.	Increases operational costs and raises concerns regarding sustainable AI development.	Design green AI algorithms, energy-efficient model architectures, and carbon-aware training strategies.	Moderate
Limited Interdisciplinary Collaboration	Collaboration among AI researchers, atmospheric scientists, public health experts, and policymakers remains limited.	Slows the translation of research findings into practical environmental management and policy.	Foster interdisciplinary research programmes, shared data platforms, and stakeholder engagement throughout model development and deployment.	High

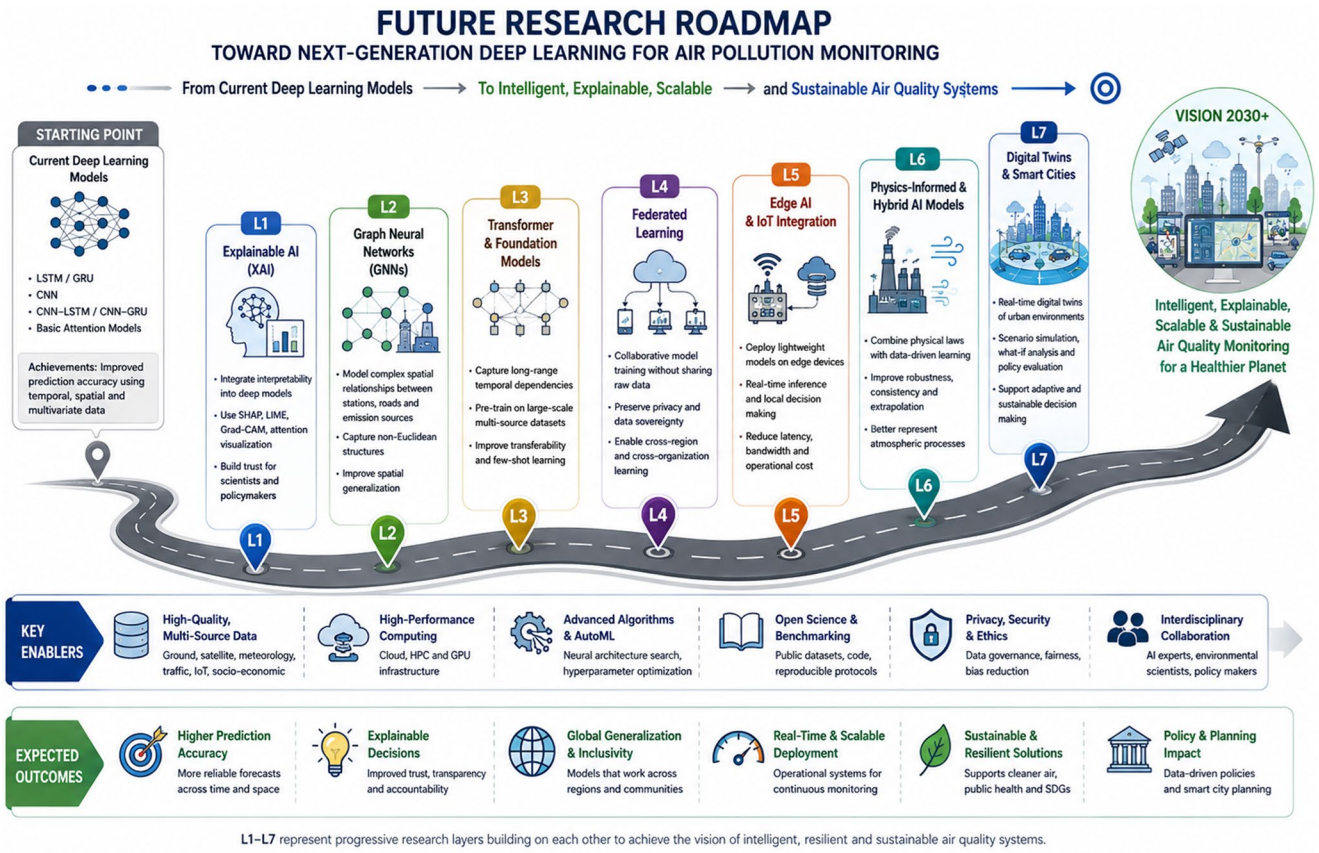


Fig. 13 Emerging research directions

5. Establish standard benchmark datasets and evaluation protocols to enable fair comparison among competing models.
6. Investigate federated, self-supervised, and physics-informed learning paradigms to improve scalability, privacy, and scientific consistency.
7. Increase research efforts in underrepresented regions, particularly Africa and other low- and middle-income countries, where air pollution monitoring infrastructure remains limited.
8. Strengthen collaboration among environmental scientists, atmospheric chemists, computer scientists, public health researchers, and policymakers to accelerate the translation of research into operational air quality management systems.

Table 10 presents a summary on future research directions and emerging technologies, potential application, expected benefits and implementation challenges as identified in literature.

In Table 11 is a recommendation for researchers, practitioners, and policymakers based on advancing deep learning-based air pollution monitoring.

6.10 Strategic Recommendations

The synthesis of the reviewed literature suggests several overarching priorities for advancing deep learning-based air pollution monitoring. First, researchers should prioritise the development of transparent, transferable, and computationally efficient models capable of operating across diverse environmental settings. Second, practitioners should focus on integrating real-time sensing technologies, multimodal data fusion, and standardised evaluation frameworks to support reliable operational deployment. Third, policymakers should strengthen monitoring infrastructure, data-sharing policies, and regulatory frameworks that facilitate the responsible adoption of artificial intelligence in environmental management.

Achieving these objectives will require sustained interdisciplinary collaboration among computer scientists, atmospheric scientists, environmental engineers, public health experts, and government agencies. Such collaboration will accelerate the development of intelligent, explainable, and scalable air quality monitoring systems that can support evidence-based policymaking, protect public health, and contribute to the achievement of sustainable development goals.

Table 10 Future research directions and emerging technologies

Emerging Technology	Potential Applications	Expected Benefits	Implementation Challenges
Explainable Artificial Intelligence (XAI)	Model interpretation, feature importance analysis, decision support for policymakers, pollution source attribution	Improves transparency, accountability, and user trust; facilitates adoption in environmental decision-making	Increased computational overhead; balancing explainability with predictive accuracy; lack of standardised XAI evaluation methods
Graph Neural Networks (GNNs)	Spatial modelling of monitoring stations, traffic networks, industrial emissions, and urban pollution dispersion	Captures complex spatial dependencies; improves urban-scale prediction accuracy; supports network-based environmental analysis	Requires graph construction expertise; computationally intensive for large networks; limited benchmark datasets
Transformer and Foundation Models	Long-term forecasting, multimodal learning, transfer learning, large-scale environmental modelling	Superior modelling of long-range temporal dependencies; enhanced scalability; improved transferability across regions	Large computational and memory requirements; extensive training data needed; high energy consumption
Federated Learning (FL)	Collaborative model training across environmental agencies, smart cities, and research institutions without sharing raw data	Preserves data privacy; enables large-scale collaborative learning; supports distributed environmental monitoring	Communication overhead; heterogeneous local datasets; synchronization and convergence challenges
Edge Artificial Intelligence (Edge AI)	Real-time air quality monitoring using IoT devices, drones, wearable sensors, and smart city infrastructure	Low latency; reduced cloud dependency; energy-efficient local inference; supports continuous monitoring	Limited hardware resources; model compression requirements; device heterogeneity; battery constraints
Self-Supervised Learning (SSL)	Representation learning from large volumes of unlabeled environmental data	Reduces dependence on labelled datasets; improves model generalisation; lowers annotation costs	Designing effective pretext tasks; limited environmental benchmarks; evaluation complexity
Semi-Supervised Learning	Air quality prediction using small labelled datasets combined with large unlabeled datasets	Improves performance where labelled observations are scarce; supports developing regions	Sensitive to pseudo-label quality; risk of propagating prediction errors
Multimodal Data Fusion	Integration of satellite imagery, meteorological data, ground stations, IoT sensors, GIS, traffic, and socioeconomic data	Enhances prediction accuracy, robustness, and spatial–temporal understanding; improves resilience to missing data	Data alignment across spatial and temporal resolutions; heterogeneous data formats; increased preprocessing complexity
Digital Twins	Virtual representation of urban environments for pollution simulation, forecasting, and decision support	Enables real-time monitoring, scenario analysis, adaptive management, and predictive maintenance of environmental systems	High implementation cost; integration of heterogeneous data streams; computational scalability
Internet of Things (IoT) and Smart Sensors	Continuous environmental monitoring, early warning systems, smart city applications	High-frequency real-time data collection; improved spatial coverage; community-based monitoring	Sensor calibration, data quality assurance, cybersecurity, communication reliability
Big Data Analytics and Cloud Computing	Large-scale environmental data processing, distributed model training, operational forecasting	Efficient handling of massive environmental datasets; scalable computation; supports collaborative research	Infrastructure costs; cloud security concerns; data governance and interoperability
Uncertainty Quantification and Bayesian Deep Learning	Probabilistic air quality forecasting, risk assessment, decision support	Provides confidence estimates; improves reliability for public health and policy decisions	Increased computational requirements; limited adoption in environmental applications
Green Artificial Intelligence	Energy-efficient model development and sustainable AI deployment	Reduces computational energy consumption and carbon footprint; promotes environmentally sustainable AI systems	Balancing efficiency with prediction accuracy; lack of standardised sustainability metrics

6.11 Summary

The literature demonstrates that deep learning has substantially advanced the forecasting, and prediction of air pollution by enabling the modelling of complex nonlinear and spatiotemporal relationships. Nevertheless, important challenges remain regarding data quality, model generalisation, interpretability, computational efficiency, and operational deployment. Emerging paradigms, including explainable artificial intelligence, graph neural networks, transformer-based architectures, federated learning and edge AI, offer promising opportunities for addressing these limitations. Future research should emphasise the development of transparent, efficient, and transferable deep learning models capable of supporting real-time environmental monitoring and evidence-based decision-making across diverse geographical contexts.

6.12 Limitations of the Review

Although this review provides a comprehensive synthesis of deep learning techniques for air pollution, forecasting,

Table 11 Recommendation for researchers, practitioners, and policymakers

Stakeholder	Key Challenge	Recommendation	Expected Outcome
Researchers	Limited availability of high-quality datasets	Develop open-access benchmark datasets integrating ground stations, satellite observations, meteorological variables, IoT sensors, and GIS information.	Improved reproducibility, model comparison, and collaborative research.
Researchers	Poor model generalisation across geographical regions	Investigate transfer learning, domain adaptation, federated learning, and foundation models for cross-regional prediction.	More robust and globally applicable prediction models.
Researchers	Black-box nature of deep learning models	Incorporate Explainable Artificial Intelligence (XAI) techniques such as SHAP, LIME, Grad-CAM, and attention visualisation.	Increased model transparency, interpretability, and stakeholder trust.
Researchers	High computational complexity	Develop lightweight deep learning architectures through pruning, quantisation, knowledge distillation, and efficient neural architecture search.	Reduced computational cost and wider deployment on resource-constrained devices.
Researchers	Limited integration of atmospheric science	Combine deep learning with atmospheric chemistry, numerical weather prediction, and physics-informed neural networks (PINNs).	More physically consistent and scientifically reliable predictions.
Researchers	Lack of uncertainty estimation	Integrate probabilistic forecasting, Bayesian deep learning, and uncertainty quantification into prediction models.	Improved confidence in decision support and risk assessment.
Researchers	Underrepresentation of developing regions	Conduct region-specific studies in Africa, South America, and other underrepresented regions through international collaborations and open data initiatives.	Greater geographical diversity and improved global applicability of deep learning models.
Practitioners (Environmental Agencies)	Limited real-time monitoring capability	Deploy IoT-enabled sensor networks integrated with Edge AI for continuous air quality monitoring and early warning systems.	Faster of pollution events and improved operational efficiency.
Practitioners	Data fragmentation across multiple platforms	Implement interoperable data management systems capable of integrating heterogeneous environmental datasets.	Better data accessibility and enhanced decision-making.
Practitioners	Difficulty selecting appropriate prediction models	Adopt standardised evaluation protocols and benchmark datasets for model comparison before operational deployment.	Improved model reliability and easier technology adoption.
Practitioners	High operational costs	Utilise cloud-edge hybrid computing architectures to optimise computational resources and reduce infrastructure costs.	Cost-effective and scalable monitoring systems.
Practitioners	Limited model maintenance	Establish continuous model retraining and monitoring pipelines using newly acquired environmental data.	Sustained prediction accuracy under changing environmental conditions.
Policymakers	Insufficient monitoring infrastructure	Increase investment in national air quality monitoring networks, particularly in underserved urban and rural regions.	Improved environmental surveillance and public health protection.
Policymakers	Limited access to environmental data	Promote open-data policies that encourage data sharing among research institutions, environmental agencies, and governments.	Accelerated scientific innovation and evidence-based policymaking.
Policymakers	Weak regulatory support for AI adoption	Develop standards and regulatory frameworks governing the responsible use of artificial intelligence in environmental monitoring.	Safe, transparent, and ethical deployment of AI technologies.
Policymakers	Fragmented institutional collaboration	Encourage partnerships among academia, industry, government agencies, and international organizations to address air pollution collaboratively.	Enhanced knowledge exchange, technology transfer, and coordinated pollution management.
Policymakers	Climate change and increasing pollution risks	Integrate AI-driven air quality forecasting into national climate adaptation and public health strategies.	More proactive environmental management and improved resilience to pollution events.
All Stakeholders	Need for sustainable AI deployment	Promote Green AI practices by encouraging energy-efficient model development, renewable-powered data centres, and environmentally sustainable computing.	Reduced environmental footprint of artificial intelligence and sustainable long-term deployment.

and prediction, several limitations should be acknowledged. The rapid evolution of artificial intelligence and deep learning presents an inherent limitation for review studies. New architectures, learning paradigms, benchmark datasets, and computational frameworks continue to emerge at a rapid pace. As a result, some recently developed techniques or applications may not be fully represented in the literature included in this review.

Also, considerable heterogeneity exists among the reviewed studies regarding datasets, pollutant types,

geographical coverage, forecasting horizons, preprocessing methods, evaluation metrics, and experimental protocols. This diversity makes direct quantitative comparison of model performance challenging and limits the ability to identify a universally superior deep learning architecture. Consequently, the comparative analysis presented in this review should be interpreted within the context of the specific datasets and application scenarios investigated in the original studies.

Finally, although this review examines a wide range of environmental data sources, including ground monitoring stations, satellite remote sensing, meteorological observations, IoT sensors, and multi-source data fusion, the availability and quality of these datasets vary substantially across different geographical regions. In particular, the limited availability of long-term monitoring data in many low- and middle-income countries, especially across Africa, restricts the generalisability of existing findings and highlights an important gap in the current body of knowledge.

7 Conclusion

This review presented a comprehensive synthesis of recent advances in deep learning techniques for air pollution forecasting and prediction. The work demonstrates that deep learning has substantially advanced air quality monitoring by enabling the modelling of complex nonlinear and spatiotemporal relationships that are difficult to capture using conventional statistical and machine learning approaches. CNNs have proven highly effective for spatial feature extraction and pollution source while LSTM and GRU models remain the dominant architectures for temporal forecasting. Transformer-based and hybrid models have achieved state-of-the-art performance by integrating spatial and temporal learning capabilities and effectively exploiting heterogeneous environmental datasets.

The findings also highlight that model performance depends not only on the choice of deep learning architecture but also on the quality and diversity of input data. Ground-based monitoring stations continue to provide the most reliable measurements, whereas satellite remote sensing, meteorological observations and IoT-based sensors significantly enhance spatial and temporal coverage. The increasing adoption of multi-source data fusion has consistently improved prediction accuracy, robustness, and model generalisation, making it a defining trend in contemporary air pollution research.

Despite these advances, several challenges like data scarcity, inconsistent monitoring infrastructure, limited model interpretability, high computational requirements, and the lack of standardised benchmark datasets continue to hinder the operational deployment of deep learning models.

Looking ahead, emerging technologies such as Explainable Artificial Intelligence, Graph Neural Networks, Physics-Informed Neural Networks, Federated Learning, Edge AI, and multimodal data fusion are expected to redefine the next generation of intelligent air quality monitoring systems.

Overall, this review provides a structured and up-to-date synthesis of the current state of deep learning for air pollution monitoring and offers practical insights for researchers, practitioners, and policymakers. By identifying existing knowledge gaps, evaluating current methodologies, and outlining future research priorities, the review establishes a comprehensive reference for advancing intelligent air quality monitoring systems. Continued interdisciplinary collaboration, together with the development of transparent, efficient, and scalable deep learning solutions, will be essential for addressing future air quality challenges and supporting healthier, more sustainable communities.

Acknowledgements The authors would like to express their sincere gratitude to all individuals and institutions that provided support, guidance, and resources throughout the course of this study. Their contributions were invaluable to the successful completion of this research.

Author Contributions E.K.: Conceptualization, methodology, literature review, data curation, manuscript drafting, and original draft preparation and corresponding author responsibilities. M.A.Y.: Literature review, formal analysis, data collection, validation, manuscript writing, and editing. T.R.: Supervision, conceptualization, methodology design, critical revision of the manuscript, project administration. K.A.A.: Data analysis, visualization, manuscript review, and technical validation. M.K.B.: Literature survey, data interpretation, manuscript editing, and validation. S.D.O.: Investigation, resource provision, manuscript review, and technical support. S.K.K.: Formal analysis, proofreading, manuscript editing, and final approval of the submitted version. All authors contributed significantly to the study, reviewed the manuscript critically, approved the final version, and agreed to be accountable for all aspects of the work.

Funding This research received no external funding. The authors did not receive any financial support from public, commercial, or not-for-profit funding agencies for the conduct of this study, the preparation of the manuscript, or its publication.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Open research No new data and software were created or analysed in this study. Data sharing is not applicable to this article.

Competing Interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

1. Kumar G, Kumar S, Suman (2020) Air quality index – A comparative study for assessing the status of air quality before and after lockdown for Meerut, in materials today: Proceedings, pp. 3497–3500. https://doi.org/10.1016/j.matpr.2021.05.575
2. Anggraini TS, Irie H, Sakti AD, Wikantika K (2024) Machine learning-based global air quality index development using remote sensing and ground-based stations. Environ Adv 15. https://doi.org/10.1016/j.envadv.2023.100456
3. Kainat S et al (2025) Sources of Air Pollution in Relation To the Public Health and Envoironment. Kashf J Multidiscip Res 2(12):62–92. https://doi.org/10.71146/kjmr793
4. WHO (2021) global air quality guidelines: particulate matter ($PM_{2,5}$ and PM_{10}), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide. Executive Summary
5. Glencross DA, Ho TR, Camiña N, Hawrylowicz CM, Pfeffer PE (2020) Air pollution and its effects on the immune system. https://doi.org/10.1016/j.freeradbiomed.2020.01.179
6. Shahraiyni HT, Sodoudi S (2016) Statistical modeling approaches for PM10 prediction in urban areas; a review of 21st-century studies. https://doi.org/10.3390/atmos7020015
7. Kumar P et al (2015) The Rise of Low-cost Sensing for managing Air Pollution in Cities. Environ Int 75:199–205. https://doi.org/10.1016/j.envint.2014.11.019
8. Bagkis E et al (2025) Evolving trends in application of low-cost air quality sensor networks: challenges and future directions. npj Clim Atmos Sci 8(1). https://doi.org/10.1038/s41612-025-01216-4
9. Amegah AK (2018) Proliferation of low-cost sensors. What prospects for air pollution epidemiologic research in Sub-Saharan Africa? Environ Pollut 241:1132–1137. https://doi.org/10.1016/j.envpol.2018.06.044
10. Van Donkelaar A et al (2016) Global Estimates of Fine Particulate Matter using a Combined Geophysical-Statistical Method with Information from Satellites, Models, and Monitors. Environ Sci Technol 50(7):3762–3772. https://doi.org/10.1021/acs.est.5b05833
11. Shi L et al (2021) A national cohort study (2000–2018) of long-term air pollution exposure and incident dementia in older adults in the United States. Nat Commun 12(1). https://doi.org/10.1038/s41467-021-27049-2
12. Hammer MS et al (2020) Global Estimates and Long-Term Trends of Fine Particulate Matter Concentrations (1998–2018). Environ Sci Technol 54:7879–7890. https://doi.org/10.1021/acs.est.0c01764
13. Li X, Peng L, Hu Y, Shao J, Chi T (2016) Deep learning architecture for air quality predictions. Environ Sci Pollut Res 23(22). https://doi.org/10.1007/s11356-016-7812-9
14. Li S, Tang B, Deng X (2025) A hybrid method combining variational mode decomposition and deep neural networks for predicting PM2.5 concentration in China, IEEE Access, vol. 13, no. March, pp. 51956–51968. https://doi.org/10.1109/ACCESS.2025.3552229
15. Kotei E, Thirunavukarasu R (2024) A Comprehensive Review on Advancement in Deep Learning Techniques for Automatic Detection of Tuberculosis from Chest X-ray Images. Springer Sci Bus Media B V https://doi.org/10.1007/s11831-023-09987-w
16. Sawah MS (2025) A novel feature selection-driven framework for sustainable and efficient rainfall classification using machine learning models, Unconv Resour vol. 9, no. October p. 100286, 2026. https://doi.org/10.1016/j.uncres.2025.100286
17. Tello-Leal E, Ramirez-Alcocer UM, Macías-Hernández BA, Hernandez-Resendiz JD (2024) Evaluation of Deep Learning Models for Predicting the Concentration of Air Pollutants in Urban Environments. Sustain 16(16):1–19. https://doi.org/10.3390/su16167062
18. Aljuaid H, Alwabel N (2019) Air pollution prediction using machine learning algorithms, Int J Eng Adv Technol. 2019;8(6 Suppl 3):160–164. https://doi.org/10.35940/ijeat.F1026.0986S319
19. Khan AA, Majid MA, Dandoush A (2025) Int J Adv Comput Sci Appl 16(3):1177–1185. https://doi.org/10.14569/IJACSA.2025.01603113. Image-Based Air Quality Estimation Using Convolutional Neural Network Optimized by Genetic Algorithms: A Multi-Dataset Approach
20. Chang YS, Chiao HT, Abimannan S, Huang YP, Tsai YT, Lin KM (2020) An LSTM-based aggregated model for air pollution forecasting. Atmos Pollut Res 11(8):1451–1463. https://doi.org/10.1016/j.apr.2020.05.015
21. Xayasouk T, Lee H, Lee G (2020) Air pollution prediction using long short-term memory (LSTM) and deep autoencoder. (DAE) Models
22. Ye T, Han M (2025) Indoor Air Pollution Source Localization Based on Small-Sample Training Convolutional Neural Networks. Buildings 15(8). https://doi.org/10.3390/buildings15081244
23. Kowalski PA, Sapała K, Warchałowski W (2020) PM10 Forecasting through applying Convolution Neural Network Techniques. Int J Environ Impacts 3(1):31–43. https://doi.org/10.2495/EI-V3-N1-31-43
24. Sawah MS, Taie SA, Ibrahim MH, Hussein SA (2023) An accurate traffic flow prediction using long-short term memory and gated recurrent unit networks. Bull Electr Eng Inf 12(3):1806–1816. https://doi.org/10.11591/eei.v12i3.5080
25. Sawah MS, Elmannai H, El-Bary AA, Lotfy K, Sheta OE (2025) Improving air quality prediction using hybrid BPSO with BWAO for feature selection and hyperparameters optimization. Sci Rep 15(1):1–20. https://doi.org/10.1038/s41598-025-95983-y
26. Athira V, Geetha P, Vinayakumar R, Soman KP (2018) DeepAirNet: Applying Recurrent Networks for Air Quality Prediction. Procedia Comput Sci 132:1394–1403. https://doi.org/10.1016/j.procs.2018.05.068
27. Lei TMT et al (2025) Application of Deep Learning Techniques for Air Quality Prediction: A Case Study in Macau. Processes 13(5):1–18. https://doi.org/10.3390/pr13051507
28. Balaraman S, Pachaivannan P, Elamparithi PN, Manimozhi S (2022) Application of LSTM models in predicting particulate matter (PM2.5) levels for urban area. J Eng Res 10(3):71–90. https://doi.org/10.36909/jer.11781
29. Sun H et al (2022) Development of an LSTM broadcasting deep-learning framework for regional air pollution forecast improvement. Geosci Model Dev 15:8439–8452. https://doi.org/10.5194/gmd-15-8439-2022
30. Kaveh M, Mesgari MS, Kaveh M (2025) A novel evolutionary deep learning approach for PM2.5 prediction using remote

sensing and spatial–temporal data: a case study of Tehran. ISPRS Int J Geo-Information 14(2). https://doi.org/10.3390/ijgi14020042
31. Zhang B et al (2023) A spatial correlation prediction model of urban PM2.5 concentration based on deconvolution and LSTM. Neurocomputing 544:126280. https://doi.org/10.1016/j.neucom.2023.126280
32. Ghajari YE, Kaveh M, Martín D (2023) Predicting PM10 concentrations using evolutionary deep neural network and satellite-derived aerosol optical depth. Mathematics 11(19). https://doi.org/10.3390/math11194145
33. Saravana Ram R, Venkatachalam K, Masud M, Abouhawwash M (2022) Air Pollution Prediction Using Dual Graph Convolution LSTM Technique. Intell Autom Soft Comput 33(3):1639–1652. https://doi.org/10.32604/iasc.2022.023962
34. Guo W, Hu Z, Jin L, Xu Y, Gonzalez MC (2025) DeepAir: deep learning and satellite imagery to estimate high-resolution PM2.5 at scale. Mach Learn Sci Technol 6(1). https://doi.org/10.1088/2632-2153/adb67a
35. Al-Janabi S, Mohammad M, Al-Sultan A (2020) A new method for prediction of air pollution based on intelligent computation. Soft Comput 24(1):661–680. https://doi.org/10.1007/s00500-019-04495-1
36. Chang W, Chen X, He Z, Zhou S (2023) A prediction hybrid framework for air quality integrated with W-BiLSTM(PSO)-GRU and XGBoost methods. Sustain 15(22). https://doi.org/10.3390/su152216064
37. Liu B, Qi Z, Gao L (2024) Enhanced Air Quality Prediction through Spatio-temporal Feature Sxtraction and Fusion: A Self-tuning Hybrid Approach with GCN and GRU. Water Air Soil Pollut 235(8):1–22. https://doi.org/10.1007/s11270-024-07346-4
38. Rejini K, Visumathi J, Genitha CH (2025) Application of transformer-based deep learning models for predicting the suitability of water for agricultural purposes. Water (Switzerland) 17(9). https://doi.org/10.3390/w17091347
39. Yilmaz M, Cinar E, Yazici A (2025) A transformer-based model for state of charge estimation of electric vehicle batteries, IEEE Access, vol. 13, no. January, pp. 33035–33048. https://doi.org/10.1109/ACCESS.2025.3542961
40. Liang Y et al (2023) AirFormer: Predicting Nationwide Air Quality in China with Transformers, in Proceedings of the 37th AAAI Conference on Artificial Intelligence, AAAI. pp. 14329–14337. https://doi.org/10.1609/aaai.v37i12.26676
41. Cao W, Qi W, Lu P (2024) Air quality prediction based on time series decomposition and convolutional sparse self-attention mechanism transformer model, IEEE Access, vol. 12, no. October, pp. 155340–155350. https://doi.org/10.1109/ACCESS.2024.3484579
42. Liu S, Hu Y (2025) Air quality prediction based on factor analysis combined with transformer and CNN-BILSTM-ATTENTION models. Sci Rep 15(1). https://doi.org/10.1038/s41598-025-03780-4
43. Zhang Z, Zhang S (2023) Modeling air quality PM2.5 forecasting using deep sparse attention-based transformer networks. Int J Environ Sci Technol 20(12):13535–13550. https://doi.org/10.1007/s13762-023-04900-1
44. Yadav N et al (2023) Using deep transfer learning and satellite imagery to estimate urban air quality in data-poor regions, environ. Pollut. vol. 342, no. August p. 122914, 2024. https://doi.org/10.1016/j.envpol.2023.122914
45. Poelzl M, Kern R, Kecorius S, Lovrić M (2025) Exploration of transfer learning techniques for the prediction of PM10. Sci Rep 15(1):1–13. https://doi.org/10.1038/s41598-025-86550-6
46. Hao Y, Luo C, Li T, Zhang J, Chen H (2025) Adversarial transfer learning-based hybrid recurrent network for air quality prediction. Int J Intell Syst. vol. no. 1, 2025. https://doi.org/10.1155/int/6014262
47. Sonawani S, Patil K (2024) Air quality measurement, prediction and warning using transfer learning based IOT system for ambient assisted living. Int J Pervasive Comput Commun 20(1):38–55. https://doi.org/10.1108/IJPCC-07-2022-0271
48. Dutta D, Pal SK (2023) Prediction and assessment of the impact of COVID-19 lockdown on air quality over Kolkata: a deep transfer learning approach. Springer International Publishing. 195(1)https://doi.org/10.1007/s10661-022-10761-x
49. Ma Z et al (2024) Air pollutant prediction model based on transfer learning two-stage attention mechanism. Sci Rep 14(1):1–18. https://doi.org/10.1038/s41598-024-57784-7
50. Yin C, Li W, Li T, Yuan S, Tang D, Dong X (2026) Air quality prediction model based on deep learning hybrid framework. Sci Rep 16(1):7084. https://doi.org/10.1038/s41598-026-37896-y
51. Iskandaryan D, Ramos F, Trilles S (2022) Graph neural network for air quality prediction: a case study in madrid, IEEE Access, vol. 11, no. November pp. 2729–2742, 2023. https://doi.org/10.1109/ACCESS.2023.3234214
52. Phung TH, Nguyen DL, Vu VH, Huynh TT, Nguyen TH, Nguyen PL (2022) Unsupervised air quality interpolation with attentive graph neural network, in ACM international conference proceeding series, Association for Computing Machinery, pp. 103–110. https://doi.org/10.1145/3568562.3568657
53. Chabalala V et al (2026) Spatiotemporal graph neural networks for PM2.5 concentration forecasting. Air 4(1). https://doi.org/10.3390/air4010002
54. Yogapriya J, Deepa S, Radha N, Madhumitha E (2024) Graph-based neural network model for predicting urban environmental air quality using spatio-temporal data optimization. Glob Nest J 26(2). https://doi.org/10.30955/gnj.005598
55. Iskandaryan D, Ramos F, Trilles S (2023) Graph Neural Network for Air Quality Prediction: A Case Study in Madrid. IEEE Access 11(2):2729–2742. https://doi.org/10.1109/ACCESS.2023.3234214
56. Basir NI, Tan KK, Djarum DH, Ahmad Z, Vo Dvn, Zhang Jie (2025) Autoencoder artificial neural network model for air pollution index prediction. IIUM Eng J 26(1):1–21. https://doi.org/10.31436/IIUMEJ.V26I1.2818
57. Ismael DS, Masseran N, Abu Bakar S (2026) Air pollution prediction based on stacked deep autoencoder network model. Electron 15(13):2756. https://doi.org/10.3390/electronics15132756
58. Rodríguez-García MI, Carrasco-García MG, Cubillas Fernández PR, Rodrigues Ribeiro M, Cardoso PJS, Turias IJ (2025) Air pollution forecasting using autoencoders: a classification-based prediction of NO2, PM10, and SO2 concentrations, Nitrogen (Switzerland), 6(4):1–16. https://doi.org/10.3390/nitrogen6040101
59. Le DD et al (2022) Insights into Multi-Model Federated Learning: An Advanced Approach for Air Quality Index Forecasting. Algorithms 15(11):1–20. https://doi.org/10.3390/a15110434
60. Beriwal S, Ayeelyan J (2025) Decoding pollution: a federated learning-based pollution prediction study with health ramifications using causal inferences. Electron 14(2). https://doi.org/10.3390/electronics14020350
61. Abimannan S, Assessment et al (2023) Sustain 15(18):1–34. https://doi.org/10.3390/su151813951
62. AlShehhi A, Welsch R (2023) Artificial Intelligence for improving Nitrogen Dioxide forecasting of Abu Dhabi environment agency ground-based stations. J Big Data 10(1). https://doi.org/10.1186/s40537-023-00754-z
63. Cabaneros SM, Calautit JK, Hughes B (2020) Spatial estimation of outdoor NO2 levels in Central London using deep neural networks and a wavelet decomposition technique. Ecol Modell 424. https://doi.org/10.1016/j.ecolmodel.2020.109017

64. Cai P, Zhang C, Chai J (2023) Forecasting hourly PM2.5 concentrations based on decomposition-ensemble-reconstruction framework incorporating deep learning algorithms. Data Sci Manag 6(1):46–54. https://doi.org/10.1016/j.dsm.2023.02.002
65. Chang YS, Abimannan S, Chiao HT, Lin CY, Huang YP (2020) An Ensemble Learning Based Hybrid Model and Framework for Air Pollution Forecasting. Environ Sci Pollut Res 27(30):38155–38168. https://doi.org/10.1007/s11356-020-09855-1
66. Inam SA et al (2024) PR-FCNN: A data-driven hybrid approach for predicting PM2.5 concentration. Discov Artif Intell 4(1). https://doi.org/10.1007/s44163-024-00184-7
67. Li D, Liu J, Zhao Y (2022) Prediction of multi-site $PM_{2.5}$ concentrations in Beijing using CNN-Bi LSTM with CBAM. Atmos 13(10):1719. https://doi.org/10.3390/atmos13101719
68. Rowley A, Karakuş O (2023) Predicting air quality via multimodal AI and satellite imagery. Remote Sens Environ 293:113609. https://doi.org/10.1016/j.rse.2023.113609
69. Holloway T et al (2021) Satellite Monitoring for Air Quality and Health. Annu Rev Biomed Data Sci 4:417–447. https://doi.org/10.1146/annurev-biodatasci-110920-093120
70. Jain V et al (2023) A hybrid approach for integrating micro-satellite images and sensors network-based ground measurements using deep learning for high-resolution prediction of fine particulate matter (PM2.5) over an indian city, lucknow, Atmos Environ vol. 338, no. December p. 120798, 2024. https://doi.org/10.1016/j.atmosenv.2024.120798
71. Sohrabi Z, Maleki J (2025) Fusing satellite imagery and ground-based observations for PM2.5 air pollution modeling in Iran using a deep learning approach. Sci Rep 15(1):1–16. https://doi.org/10.1038/s41598-025-05332-2
72. Muruganandam NS, Arumugam U (2022) Seminal stacked long short-term memory (SS-LSTM) model for forecasting particulate matter (PM2.5 and PM10). Atmos 13(10):1726. https://doi.org/10.3390/atmos13101726
73. Rad AK, Nematollahi MJ, Pak A, Mahmoudi M (2025) Predictive modeling of air quality in the Tehran megacity via deep learning techniques. Sci Rep 15(1):1–20. https://doi.org/10.1038/s41598-024-84550-6
74. Wang B, Kong W, Guan H, Xiong NN (2019) Air Quality Forecasting Based on Gated Recurrent Long Short Term Memory Model in Internet of Things. IEEE Access 7:69524–69534. https://doi.org/10.1109/ACCESS.2019.2917277
75. Ayele TW, Mehta R (2018) Air pollution monitoring and prediction using IoT, in Proceedings of the International Conference on Inventive Communication and Computational Technologies, ICICCT IEEE pp. 1741–1745. https://doi.org/10.1109/ICICCT.2018.8473272
76. Ramadan MNA, Ali MAH, Khoo SY, Alkhedher M, Alherbawi M (2024) Real-time IoT-powered AI system for monitoring and forecasting of air pollution in industrial environment. Ecotoxicol Environ Saf 283:116856. https://doi.org/10.1016/j.ecoenv.2024.116856
77. Banciu C, Florea A, Bogdan R (2024) Monitoring and predicting air quality with IoT devices. Processes 12(9). https://doi.org/10.3390/pr12091961
78. Moursi AS, El-Fishawy N, Djahel S, Shouman MA (2021) An IoT enabled system for enhanced air quality monitoring and prediction on the edge. Complex Intell Syst 7(6):2923–2947. https://doi.org/10.1007/s40747-021-00476-w
79. Kalaiselvi S, Anitha V, Manimaran V, Lawrence TS (2025) Air quality prediction using multi-source remote sensing data integration with hybrid deep learning framework. Sci Rep pp.1–10. https://doi.org/10.1038/s41598-025-32466-0
80. Liu Y, Chang HH, Fu JS (2026) Data fusion in air pollution exposure assessment: Methods, applications, and future directions. J Air Waste Manag Assoc 76(6):423–462. https://doi.org/10.1080/10962247.2026.2660605
81. Bhatti MA, Song Z, Bhatti UA, Syam MS (2024) AIoT-driven multi-source sensor emission monitoring and forecasting using multi-source sensor integration with reduced noise series decomposition. J Cloud Comput 13(1). https://doi.org/10.1186/s13677-024-00598-9
82. Raza T, Singh K (2025) AI-Driven multisource data fusion for real-time urban air quality forecasting and health risk assessment. J Sci Innov Adv Res. 1(2). [Online]. Available: https://jsiar.com
83. He Z, Guo Q (2024) Comparative analysis of multiple deep learning models for forecasting monthly ambient PM2.5 concentrations: a case study in Dezhou City, China. Atmos (Basel) 15(12). https://doi.org/10.3390/atmos15121432
84. Naz F et al (2023) Comparative Analysis of Deep Learning and Statistical Models for Air Pollutants Prediction in Urban Areas. IEEE Access 11:64016–64025. https://doi.org/10.1109/ACCESS.2023.3289153
85. Chakraborty S, Dipto SM, Pilkiewicz KR, Mayo ML, Rana P (2026) A Comparative Analysis of Explainable AI (XAI) Techniques for Transparent and Reliable Image Classification. Entropy 28(5):1–19. https://doi.org/10.3390/e28050562
86. Imam NM, Ibrahim A, Tiwari M (2024) Explainable Artificial Intelligence (XAI) Techniques To Enhance Transparency In Deep Learning Models. IOSR J Comput Eng 26(6):29–36. https://doi.org/10.9790/0661-2606012936

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.